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Review article



Past, present and future of deep plant leaf disease recognition: A survey

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ABSTRACT

Agriculture is the foundation of life that faces numerous daily attacks from nature and living organisms. A major challenge for farmers is timely plant disease identification, which is crucial to prevent productivity losses and the production of poor-quality products. Researchers have recently been focusing on automating the plant leaf disease recognition process using computer vision and machine learning techniques. More importantly, the recent developments in deep learning have significantly advanced the field of plant leaf disease recognition. Regardless of these advancements, significant challenges remain in automatic leaf disease recognition, and researchers are continuing to seek better performance, in-field applicability, and compatibility with resource-constrained devices. This survey provides a comprehensive overview of real-world and laboratory datasets, feature extraction methods, deep learning frameworks, limitations, recommendations, and future directions for deep plant leaf disease recognition. It offers a detailed comparative analysis of various deep learning models applied to different datasets, preprocessing techniques, and data collection methods. This work also highlights the need for an ideal dataset and explores future directions like the Internet of Things integration, Explainable AI, and Smart Farming, which previous surveys have not covered. The primary aim of this survey is to assist researchers in understanding state-of-the-art plant leaf disease recognition techniques, support farmers in the field of plant pathology, address limitations, provide recommendations and outline future directions.

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1. Introduction

Agriculture is the backbone of human civilization and serves as the primary source of food supply. The growing demand for food, driven by an increasing population, challenges farmers and agricultural scientists, who desperately seek immediate and sustainable solutions. In recent years, every country has been trying to modernize its agricultural practices in various ways to improve overall production and sustainability (King, 2017). For instance, rice quality evaluation systems can now accurately analyze grain size, shape, color, and texture, facilitating more reliable and objective measurements of rice quality (Yang et al., 2021). Similarly, automated fruit inspection techniques can assess ripeness, detect bruising, and identify defects in real-time, ensuring that only high-quality produce reaches the market (Zhang et al., 2024b). Furthermore, automated plant leaf disease recognition technologies have emerged, enhancing the ability to efficiently identify and diagnose plant diseases, thereby supporting improved agricultural outcomes. Additionally, these technologies contribute to better land and water management, as well as other agricultural practices, fostering sustainability in the agricultural sector (Hu et al., 2022). However, the emergence of various types of infectious plant diseases caused by

pathogens has become a major threat to crops, responsible for severe food shortages and significant economic losses (Hungilo et al., 2019). Due to the severity of most plant diseases, the quality of food is also significantly diminished. Savary et al. (2019) found that the severity of plant diseases could result in yield losses ranging from 10% to 40%, depending on the crop and disease. Furthermore, plant diseases can compromise the nutritional content of crops. Sharma et al. (2019) highlighted that specific plant diseases can cause the deficiency of vital nutrients like vitamins and minerals in crops, amplifying concerns regarding food quality.

To address the challenges posed by plant diseases, researchers in the agricultural field have employed a diverse array of plant recognition techniques. Traditionally, farmers and agricultural technicians have relied on visual inspection for the identification of plant diseases. Limited knowledge, high costs, and inaccuracies are significant factors that negatively affect the manual assessment of plant diseases (Bock et al., 2010). Therefore, it is crucial to implement computer-aided plant disease recognition techniques to effectively address this problem (Savary et al., 2012). Over the years, computer vision-based techniques with machine learning (ML) have demonstrated substantial advancements in accurately recognizing plant diseases from leaf images. In earlier studies, handcrafted feature extraction techniques, which acquire low-level features, were extensively utilized. For example, in Elangovan and Nalini (2017), image segmentation and SVM techniques are utilized for feature extraction for plant leaf diseases. In another work, the fractional-order Zernike moment feature is used with support vector machine (SVM) to classify plant diseases (Kaur et al., 2019). These kinds of handcrafted feature extraction methods are constrained by the requirement for extensive knowledge of plant diseases. Furthermore, these methods are not end-to-end trainable, hindering their straightforward usage in practical applications.

Very recently, DL-based approaches have gained popularity in various tasks in many domains, demonstrating superior performance compared to traditional handcrafted feature extraction-based classification methods (Tan et al., 2021). The CNN is one of the most widely used DL architectures, especially prominent in tasks involving image classification (Garcia-Garcia et al., 2017). Specifically, CNN-based DL frameworks effectively mitigate the limitations inherent in handcrafted feature extraction techniques for plant disease recognition. For example, the AlexNet model with eight layers has exceeded the performance of existing handcrafted techniques in the recognition of plant leaf diseases (Singh et al., 2022b). Similarly, state-of-the-art deep networks, such as VGG-16 and VGG-19, are employed for recognizing plant leaf diseases (Kavala and Pothuraju, 2022). MobileNet, a notable lightweight network, frequently exhibits superior accuracy compared to existing traditional handcrafted techniques (Liu et al., 2023). Consequently, recent literature indicates that the increasing prevalence of DL-based techniques, as evidenced by the widespread adoption of advanced DL architectures, federated learning, quantum DL, and explainable AI, represents a significant paradigm shift in the domain of plant leaf disease recognition (Mensah et al., 2023; Singh and Khan, 2023; Natarajan et al., 2024).

1.1. Previous surveys

Due to the rapid evolution of plant leaf disease recognition, numerous works have been published recently. In addition to numerous original research papers, several survey papers have also been published on this topic. The strengths and limitations of each existing survey paper are evaluated, assessing whether the article provides a summary of plant stress, analyzes a wide range of publicly available datasets, includes instructions on setting up an ideal dataset, utilizes DL-based approaches, and covers an extensive number of significant applications and future directions. Table 1 provides a comparison of seven existing recent survey articles on the recognition of leaf diseases from 2020 to 2024. The important aspects of each of those survey papers are critically evaluated below.

- In Singh et al. (2020), various imaging methods for early plant disease detection that include thermal, hyperspectral, fluorescence, multispectral, and 3D imaging are reviewed. While the paper discusses challenges and trends up to 2020, it lacks detailed descriptions of *biotic* and *abiotic* plant stress factors and does not include datasets specific to plant diseases.
- In Orchi et al. (2021), transfer learning and DL in crop disease identification, as well as Internet of Things (IoT) enabled crop health, are explored through studies conducted from 2018 to 2021. However, it does not adequately address scalability in AI and IoT solutions and lacks strategies for overcoming data acquisition and pre-processing challenges.
- In Tugrul et al. (2022), studies that use CNNs for plant disease detection, including public datasets like PlantVillage and Kaggle, are reviewed. This survey proposed a web app with integrated DL for leaf disease detection but lacked insights into practical challenges and comparisons with other techniques in plant leaf disease recognition.
- In Ahmad et al. (2023b), approximately seventy studies on DL applications in plant disease classification and management in agriculture are reviewed. It includes an analysis of DL techniques, dataset requirements, imaging sensors and various datasets. However, there is a limited focus on plant stress, ideal dataset development, and real-world applications in plant leaf disease recognition and management.
- In Shoaib et al. (2023), advancements in plant disease detection using ML and DL from 2015 to 2022 are discussed. It highlights methods like image processing and feature extraction, noting benefits and limitations such as data availability and imaging quality. This survey briefly discussed nine datasets but did not deeply examine the limitations of these datasets or the economic and environmental impacts of these techniques.
- In Sajitha et al. (2024), a comprehensive survey on plant disease detection, focusing on datasets like PlantVillage, PlantDoc, and RoCoLe is conducted. This survey paper categorizes diseases and stresses, highlighting the need for larger benchmarked datasets, but lacks specifics on their setup. It also highlighted the transformative impact of IoT, sensors, drones, AI, and blockchain in agriculture, improving disease detection. However, dataset limitations and biases in ML/DL model training and testing were not extensively addressed.
- In Bhargava et al. (2024), different plant diseases, their types, and associated symptoms of plant stress are reviewed. It briefly discussed datasets used in plant disease detection systems but lacked detailed information. Furthermore, domain-specific applications of these detection systems were not addressed, and there was a lack of comprehensive discussion on future research directions.

1.2. Survey contributions

This survey paper provides better coverage than existing surveys in plant leaf disease recognition by offering a comprehensive and up-to-date analysis of the latest advancements to date. Unlike previous surveys, which often focus on a limited set of existing research and datasets, this survey paper provides an extensive review of a broader range of research, including shallow and DL, as well as specialized techniques tailored for plant leaf disease recognition. This survey ensures a more holistic understanding of the field by addressing the entire spectrum of approaches. In addition, the details of various algorithms and models are explored, comparing their performance, scalability, and applicability in different contexts. Emerging trends and future directions in this domain are also highlighted, providing valuable insights for researchers and practitioners. This survey paper differs considerably from existing survey papers in six important aspects, as indicated in Table 1. The key contributions of this survey are summarized as follows:

Table 1
Comparison of existing survey papers in plant disease recognition.

Authors, Year	Summary of key strengths and limitations		Included or Not Included (✓/✗)					
	Strengths	Limitations	PS	DS > 10	IDS	DL	APL > 5	FD
Singh et al. (2020)	Includes discussions on advancements in imaging technologies from 2003 to 2019, providing insights into real-world uses and the impact of these technologies on agriculture.	It briefly mentions DL approaches but lacks an in-depth look at recent advancements in DL models for plant disease detection, despite rapid developments in this area.	✓	✗	✗	✗	✗	✗
Orchi et al. (2021)	Provide a broad overview of research over the past decade on crop disease detection, covering various methodologies such as ML, DL, image processing techniques, IoT, and hyperspectral image analysis.	It lacks extensive discussion on the scalability of proposed AI and IoT solutions and does not adequately address overcoming limitations in data acquisition and pre-processing.	✓	✗	✓	✓	✗	✓
Tugrul et al. (2022)	Thoroughly review the application of different CNN models, comparing their performance on various datasets, and helping to understand which models are most suitable for specific types of plant diseases.	It lacks detailed insights into the practical challenges of implementing CNN-based models in real-world agricultural settings. The review focuses heavily on CNNs but does not offer a substantial comparison with other DL techniques used for plant leaf disease detection.	✗	✗	✗	✓	✗	✓
Ahmad et al. (2023b)	Provides an extensive review of the application of DL for plant disease diagnosis, studies sourced from prominent databases.	It lacks a detailed comparative analysis of various DL techniques or tools and does not thoroughly cover the datasets used in DL research for plant disease diagnosis.	✗	✗	✗	✓	✗	✗
Shoaib et al. (2023)	The inclusion of case studies and examples where DL models have been successfully applied in plant leaf disease detection with a thorough overview of recent advancements in DL models applied to plant disease detection.	It lacks discussion on limitations and challenges with datasets for training and testing DL models and does not extensively examine the economic and environmental impacts of using ML and DL techniques for plant disease detection.	✗	✗	✗	✓	✗	✓
Sajitha et al. (2024)	Provides a thorough review of image-based plant disease detection and classification systems, encompassing both ML and DL techniques.	Do not adequately address the limitations and biases present in the datasets used for training and testing ML or DL models	✓	✗	✗	✓	✗	✓
Bhargava et al. (2024)	Offer comparative studies of different methods, highlighting their performance metrics and effectiveness.	The review did not explore how AI models can be integrated with other technologies, such as IoT, remote sensing, and UAVs, for comprehensive plant disease management systems.	✓	✗	✗	✓	✗	✗

Abbreviations: Plant Stress (PS), Datasets (DS), Ideal Dataset (IDS), Application (APL), Future Directions (FD), Internet of Things (IoT), Unmanned Aerial Vehicle (UAV).

1. Highlights the common stress factors affecting plants and their impacts on plant health and productivity.
2. An in-depth comparative analysis of various plant disease datasets, highlighting their sizes, diversities, and qualities, is provided. Additionally, the paper includes suggestions for creating an ideal real-world dataset for future research in plant disease recognition.
3. The effectiveness of various state-of-the-art DL methods proposed to date for the recognition and diagnosis of plant leaf diseases is thoroughly examined and compared.
4. The limitations of existing DL approaches for plant leaf disease recognition are discussed, along with providing actionable recommendations to enhance their applicability and effectiveness in real-world agricultural practices.

1.3. Survey structure

Building upon the comprehensive survey contributions outlined, in this section, the structure of this paper is outlined. Specific research questions and the sections in which each of them is addressed throughout the paper are indicated, providing the reader with a structured to navigate the detailed analyses and insights presented here. This study aims to address the following research questions (RQs):

- **RQ1:** What are the common stress factors affecting plants, and how do they impact plant health and productivity?
- **RQ2:** What publicly available datasets exist for plant disease research, and how do they compare in terms of size, diversity, and quality?
- **RQ3:** How do state-of-the-art DL models contribute to the recognition and diagnosis of plant diseases, and what are the specific methods and techniques that enhance their effectiveness in this domain?
- **RQ4:** What are the practical applications of plant disease recognition methods in agriculture?
- **RQ5:** What are the limitations of existing plant leaf disease recognition approaches, and what recommendations can be made to address these limitations in the future?

Section 2 outlines the review method used for this survey study. Section 3 addresses *RQ1*, investigating the common stress factors that affect plants and their consequent impacts on plant health and productivity. In Section 4, *RQ2* is addressed by systematically analyzing publicly available datasets to recognize plant leaf disease. Section 5 provides a detailed overview of the techniques used for pre-processing, feature extraction, and classification. In Section 6, the recent DL approaches proposed for plant leaf disease recognition are reviewed, which addresses *RQ3*. Section 7 addresses *RQ4* highlights some of the key practical applications of the methods proposed in the literature for agricultural practices. Section 8 lists some limitations and Section 9 illustrates some recommendations and future directions, where *RQ5* is addressed in both sections.

2. Review method

The methodology applied in this survey follows the *Preferred Reporting Items for Systematic Reviews and Meta-Analyses* (PRISMA) (Moher

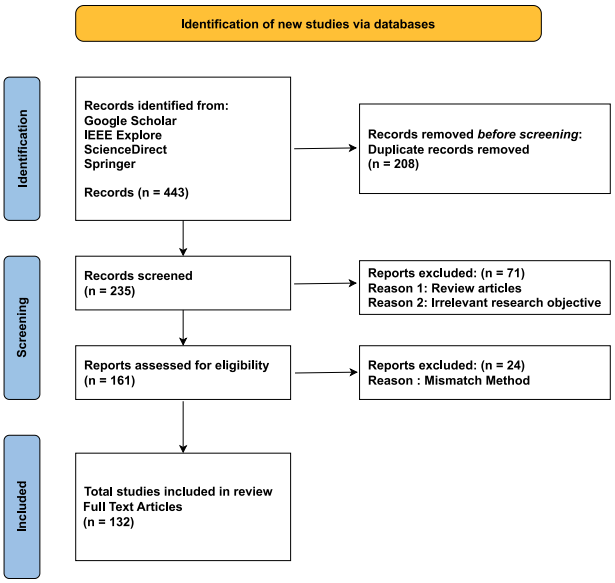


Fig. 1. The PRISMA technique-based article search strategy used in this study.

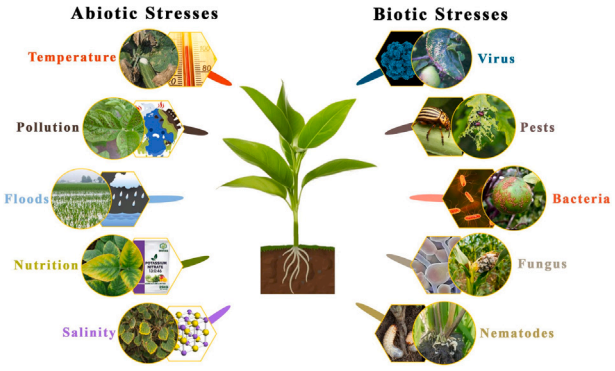


Fig. 2. Illustration of diverse impacts of *biotic* and *abiotic* stresses on plant growth across different dimensions with examples.

et al., 2009) guidelines, ensuring comprehensive, transparent, and systematic reporting of the review process, including search strategy, inclusion, and exclusion criteria, and reporting of selected articles.

2.1. Search strategy and selection criteria

Based on the requirements of this Systematic Literature Review (SLR) for plant leaf disease recognition, the search terms are: "`Plant Disease Detection`", "`Leaf Disease Recognition`", "`Plant Disease Detection`" AND "`Deep Learning`" OR "`Machine Learning`", and "`Leaf Disease Recognition`" AND "`Deep Learning`" OR "`Machine Learning`". An initial search for conference and journal articles was conducted in Google Scholar,[1] followed by a targeted search in the following digital databases: IEEE Explore,[2] Elsevier,[3] Springer[4] and ScienceDirect.[5] The search was limited to publications published between 2014 and 2024. Articles were selected based on the following criteria:

- The document is available for download in its entirety and written in English.
- The document was published between 2014 and 2024.
- The document has undergone peer review.
- The document is relevant to the research questions.
- The document presents original research or significant contributions to the field.
- The document includes empirical data or experimental results.

Before meticulously applying the search strategy and selection criteria, approximately 2,040,000 research papers relevant to the keywords had been identified in the search results. In this process, as shown in Fig. 1, 443 articles were initially identified. This subset comprehensively represents studies on plant leaf disease detection from 2014 to 2024, highlighting the growing importance of this research for global food security, sustainable agriculture, and economic stability (Devaux et al., 2020). Before the screening, 208 duplicate articles were eliminated. During the screening phase, 17 articles categorized as letters or reviews were excluded. Subsequently, we carefully reviewed the title and abstract, resulting in the identification of 57 articles with irrelevant research objectives. After a thorough examination, 161 articles remained for a detailed assessment. Finally, 29 articles were excluded because their methods did not align with the review objectives; specifically, they did not explicitly address feature extraction methods or classification algorithms. Following a sequential process of identification, screening, and eligibility testing, 132 papers were included in this review.

3. Stress on plants

Plant stress is primarily categorized as either *biotic* or *abiotic*, and it significantly impacts both the yield and quality of crop production, thereby compromising the overall quality of agricultural produce. Effectively managing stress is crucial for ensuring optimal plant productivity, resilience, and adaptability to changing environmental conditions. By implementing strategies such as genetic improvement, precision farming techniques, and sustainable agricultural practices, farmers can mitigate the adverse effects of stressors, thereby safeguarding crop yields and enhancing the overall quality of agricultural produce (Gull et al., 2019; Dresselhaus and Hückelhoven, 2018).

Fig. 2 shows two main types of stress encountered by plants: *biotic* and *abiotic*. It illustrates *abiotic* stress factors such as temperature, pollution, floods, nutrition, and salinity, as well as *biotic* stress factors including viruses, pests, bacteria, fungi, and nematodes. *Biotic* stress caused by living organisms can deplete essential nutrients in plants, potentially leading to plant death (Gull et al., 2019). On the other hand, *abiotic* stresses can damage crucial cellular structures such as cell membranes, chloroplasts, and mitochondria, altering their functionality and jeopardizing overall plant health. Additionally, *abiotic* stress induces the production of reactive oxygen species (ROS), causing oxidative stress that damages cellular components, such as deoxyribonucleic acid (DNA), proteins, and lipids, thereby exacerbating stress-induced harm (Koyro et al., 2012).

4. Plant disease datasets

Datasets play a crucial role in research and serve as foundational elements for the analysis and implementation of novel approaches. These datasets help make research more open, reproducible, and collaborative and allow researchers around the world to build on each other's work. Leaf image datasets exhibit unique characteristics such as variations in shape, size, texture, and color, along with disease-specific features like spots and lesions. These attributes contribute to improved classification performance in plant disease recognition by offering rich visual cues to distinguish healthy leaves from diseased ones (Lee et al., 2017; Ahmad et al., 2022; Al-Hiary et al., 2011;

[1] https://scholar.google.com/.
[2] https://ieeexplore.ieee.org/Xplore/home.jsp.
[3] https://www.elsevier.com/.
[4] https://www.springer.com/gp.
[5] https://www.sciencedirect.com/.

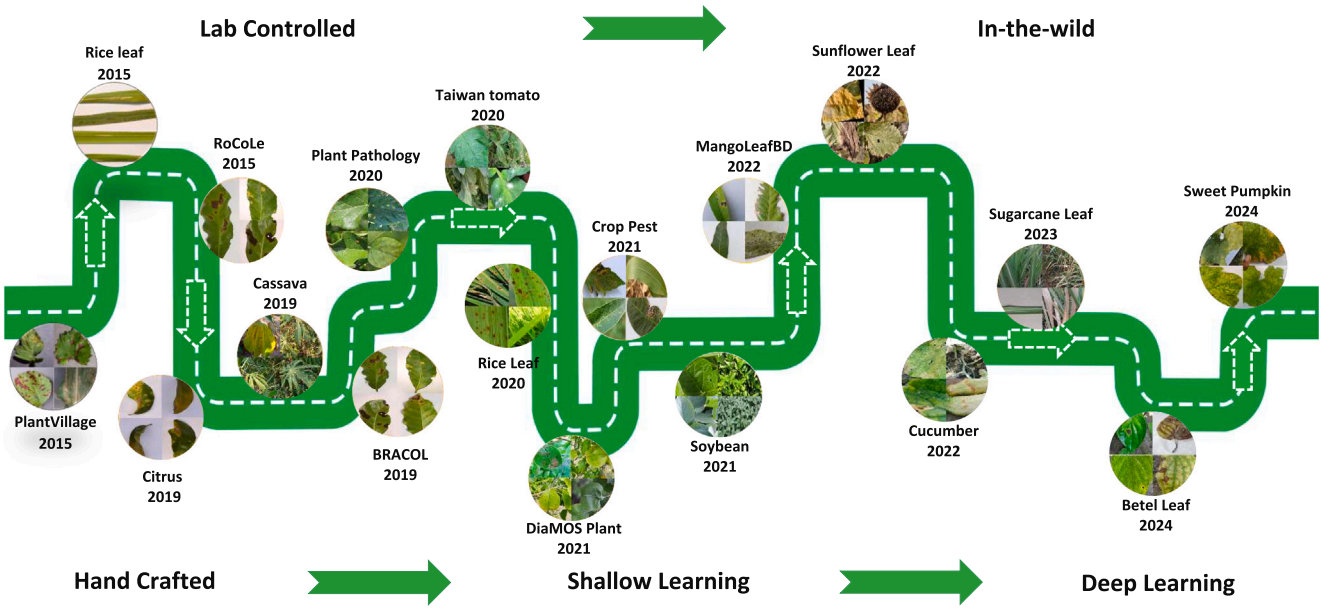


Fig. 3. Comprehensive roadmap of various plant leaf disease datasets released between 2015 and 2023.

Singh and Misra, 2017; Ahmad et al., 2021). In the field of plant leaf disease recognition, numerous public and private datasets have been used over the years to enhance the accuracy and reliability of disease recognition methods. Fig. 3 outlines a comprehensive roadmap of the various publicly available plant leaf disease datasets released between 2015 and 2023, highlighting their evolution and significance over time. As can be seen, over the years, the evolution of datasets has transformed from lab-controlled environments to in-the-wild scenarios, paralleling the shift in learning techniques from handcrafted methods to shallow and then DL approaches. Fig. 4 illustrates some sample leaf images taken from laboratory and in-wild datasets. These leaf samples demonstrate the difference between controlled laboratory conditions and the complexities of real-world agricultural environments. As shown in the figure, the complexities in real-world agricultural environments include intricate backgrounds with multiple leaves and other objects, as well as varying lighting conditions. Most of the datasets reviewed in this survey are widely recognized benchmark datasets commonly used in the research domain of plant disease detection. The plant leaf images in these datasets have been validated by plant pathology experts and agricultural scientists to ensure that each image accurately represents the corresponding disease or condition. This validation is typically carried out in collaboration with relevant institutions when the data is released by the authors. Furthermore, we ensured that the datasets analyzed in this survey are extensively utilized in research on plant diseases within the deep learning-based precision agriculture domain. For example, the PlantVillage dataset is frequently employed in studies focusing on plant health and disease detection using deep learning techniques (Kotwal et al., 2023).

4.1. Publicly available datasets

Although numerous publicly available datasets exist, this survey focuses on explaining the 18 most widely used datasets. By providing detailed descriptions and analyses of these key datasets, this survey aims to highlight their importance and utility in facilitating reproducible research. Furthermore, understanding the characteristics and applications of these datasets helps researchers select the most appropriate data. Table 2 provides comprehensive details about the 18 datasets examined in this survey. Among the 132 articles surveyed, 59.3% utilized the PlantVillage dataset, as shown in Fig. 5. The download links for these 18 datasets are provided in Table 3.

4.1.1. BRACOL

The BRACOL dataset was developed to identify and quantify coffee diseases and pests using images of Brazilian Arabica coffee leaves (Krohling et al., 2019). It comprises 1747 images, including healthy leaves and those affected by biotic stresses such as brown leaf spot, leaf miner, Cercospora leaf spot, and leaf rust. Five different smartphones were used in the image acquisition process: ASUS Zenfone 2, Xiaomi Redmi 5A, Xiaomi S2, Galaxy S8, and iPhone 6S. The images were collected in various regions of the state of Espirito Santo, Brazil, at different times of the year. The photographs were taken of the abaxial (lower) side of the leaves under partially controlled conditions with a white background. The annotation process was performed with the assistance of an expert in biotic stress recognition.

4.1.2. RoCoLe dataset

The RoCoLe dataset is composed of 1560 images of Robusta coffee leaves (Parraga-Alava et al., 2019). These images are categorized into six groups: healthy, red spider mites, and levels of rust 1 through 4. The dataset was created using a smartphone equipped with a 5 Mega Pixels (MP), f/2.2 autofocus camera, capturing images in a real-world setting at a distance of 200–300 mm, without zoom. Capture conditions include real-world factors such as varying lighting conditions, diverse backgrounds, and fluctuating temperature levels. In addition, the dataset includes annotation files for each image.

4.1.3. Plant pathology dataset

The plant pathology dataset contains 3651 images, including both healthy and unhealthy apple leaf images (Thapa et al., 2020). The dataset features images depicting apple scab, cedar apple rust, and some images of multiple diseases on the same leaf. These images were captured during the 2019 growing season from commercially grown cultivars in an unsprayed apple orchard at Cornell AgriTech in Geneva, New York, USA. Photographs were taken in real-world environments, with varying angles, illumination, surface, and noise conditions. The images were captured using a Canon Rebel T5i DSLR (Canon Inc., Tokyo, Japan) and smartphones.

4.1.4. Rice leaf diseases

The rice leaf diseases dataset was collected in the Indian village of Shertha, near Gandhinagar (Shah et al., 2019). It comprises 120 images categorized into three disease classes: bacterial leaf blight, brown spot,

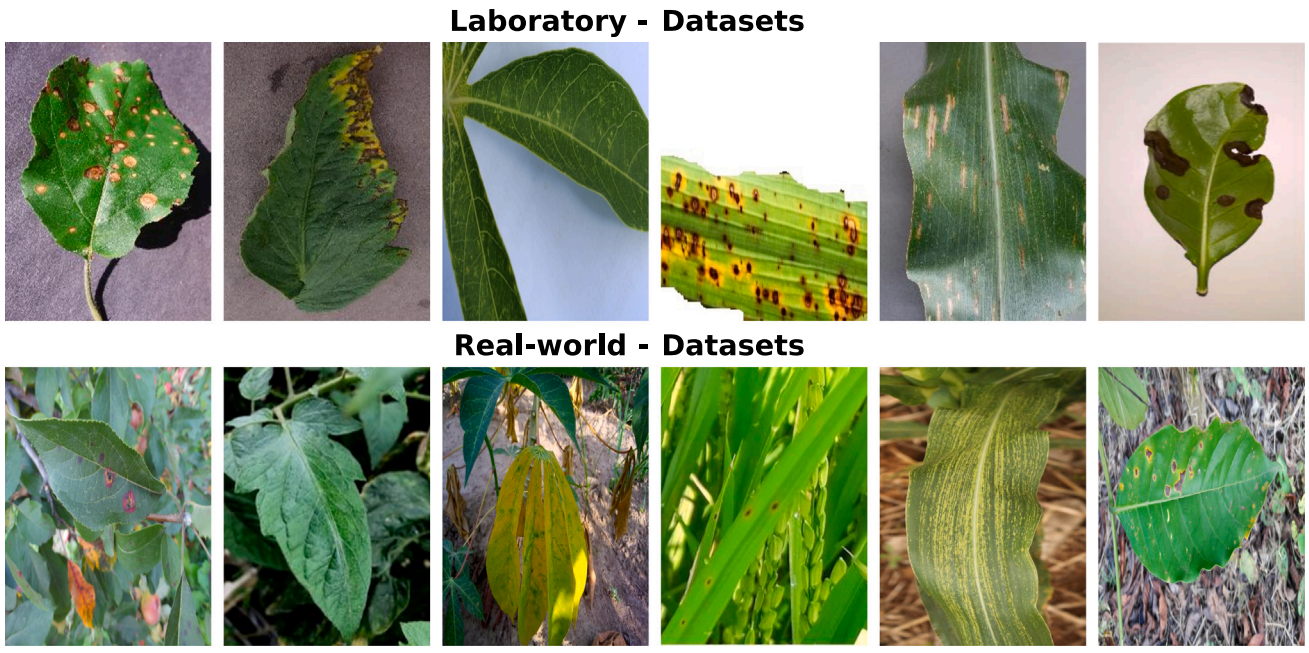


Fig. 4. Comparison of laboratory and in-the-wild plant leaf disease samples: some samples taken from various datasets.

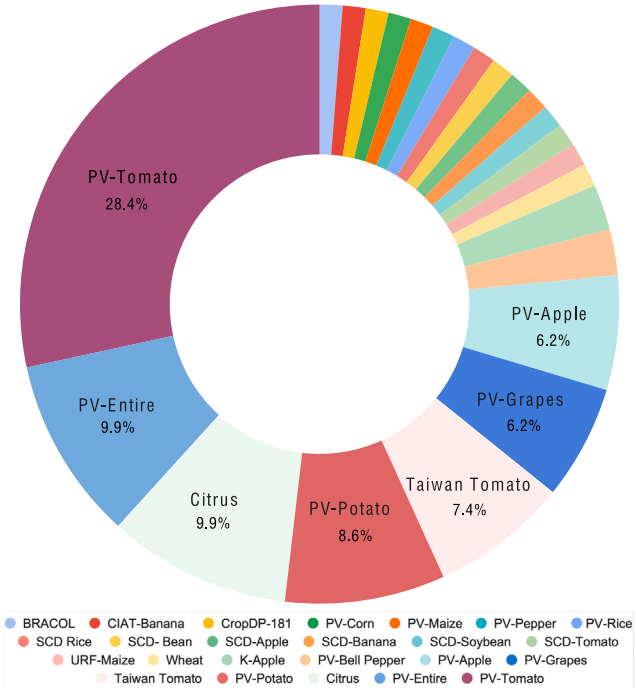


Fig. 5. Comparison of popular datasets for plant disease recognition. PV denotes PlantVillage dataset.

and leaf smut. The photographs were taken with a 12.3MP Nikon D90 digital SLR camera against a white background. Farmers assisted in the identification process in the Gujarati language, and agricultural field experts provided interpretations in English.

4.1.5. Citrus

The citrus dataset comprises 150 images of fruits and 609 images of leaves, including both healthy and unhealthy specimens (Rauf et al., 2019). For leaves, the dataset includes classes such as black spot, canker, greening, melanose, and healthy. For fruit images, additional disease classes like black spot, canker, greening, scab, and healthy are included. The images were captured using an advanced DSLR camera (Canon EOS 1300D) with a resolution of 5202 × 3465 pixels. Photographs were taken in a laboratory environment against a uniform gray background, predominantly in the Sargodha region of Pakistan in December. The images were resized to dimensions of 256 × 256 pixels with a 72 dots per inch (dpi) resolution. Data collection was carried out by citrus disease experts from the Citrus Research Center, in collaboration with officers from the Government of Punjab, Pakistan.

4.1.6. PlantVillage

The PlantVillage dataset is one of the largest datasets in plant pathology, comprising 54,390 images (Hughes et al., 2015). This extensive collection covers 14 different plant species: apple, blueberry, cherry, corn, grape, orange, peach, bell pepper, potato, raspberry, soybean, squash, strawberry, and tomato. The images were captured using a Sony DSC-RX100/13 with a resolution of 20.2MP, with each leaf represented in three different views: color, grayscale and segmented. For each leaf, 4–7 images were taken, capturing various angles with a standard point rotated around 360 degrees. The dataset encompasses images of 12 healthy crop species and 26 unhealthy crop species. Among the unhealthy images, there are 17 fungal diseases, 4 bacterial diseases, 2 mold (oomycete) diseases, 2 viral diseases, and 1 mite disease.

4.1.7. DiaMOS plant dataset

The DiaMOS Plant dataset comprises 3505 images of pear leaves and fruits, sourced from Sardegna, Italy (Fenu and Malloci, 2021). The leaf images are categorized into classes including healthy, spot, curl, and slug. The photography was carried out under various lighting conditions, such as direct sunlight, indirect sunlight, distributed light, and strong reflection of sunlight. The images were captured using a Canon EOS 60D camera and an Honor 6× smartphone, offering resolutions of 3456 × 5184 and 2976 × 3968, respectively. Captured in a real-life environment, the images feature different lighting, angles, backgrounds, and noise conditions. All images were resized to 224 × 224 × 3.

Table 2
Overview of various plant disease datasets, including information on the dataset name, number of samples, plant species, object of interest, number of classes, acquisition location, capturing device, image resolution, and color information.

Dataset	Year	Samples	Plant	Object of interest	Number of classes	Acquisition location	Capturing device	Image resolution	Color
BRACOL	2019	1747	Coffee	Leaf	5	Laboratory	Different Smartphones	Multiple	RGB
RoCoLe	2019	1560	Coffee	Leaf	6	Real-world	Smartphones	5MP	RGB
Plant Pathology	2020	3651	Apple	Leaf	4	Real-world	Canon Rebel T5i DSLR	Multiple	RGB
Rice Leaf Diseases	2017	120	Rice	Leaf	3	Laboratory	NIKON D90	12.3MP	RGB
PlantVillage	2016	54 390	14 Plants	Leaf	38	Laboratory	Sony DSC	–	RGB
Cassava Disease Leaf Dataset	–	22 031	Cassava	Leaf	5	Real-world	Smartphones	N/A	RGB
Citrus Dataset	2019	759	Citrus	Leaf, Fruits	10	Laboratory	Canon EOS 1300D	256 × 256	RGB
DiaMOS Plant Dataset	2021	3505	Pear	Leaf	5	Real-world	Smartphone & DSLR Camera	2976 × 3968 & 3456 × 5184	RGB
Taiwan Tomato Leaves	–	622	Tomato	Leaf	5	Laboratory & Real-world	N/A	N/A	RGB
Rice Leaf Diseases Dataset	–	5932	Rice	Leaf	4	Real-world	Nikon DSLR-D5600	N/A	RGB
Crop Pest and Disease Detection	2023	24 881 (10 297[a])	Cashew, Cassava, Maize & Tomato	Leaf, Pest & Fruits	22	Real-world	Canon-EOS Rebel T7	400 × 400, 487 × 1080, 1080 × 518, 3024 × 4032 & 4032 × 3024	RGB
Sugarcane Leaf Image Dataset	2023	7134	Sugarcane	Leaf	12	Laboratory & Real-world	Smartphones	72 dpi	RGB
Cucumber Disease Recognition Dataset	2022	6400	Cucumber	Fruit, Leaf	8	Real-world	–	–	RGB
MangoLeafBD Dataset	2022	4000	Mango	Leaf	7	Laboratory	Mobile phone camera	240 × 320	RGB
Betel Leaf Image Dataset	2024	3589 (2589[a])	Betel	Leaf	4	Real-world	–	–	RGB
Sunflower Leaf Dataset	2022	1668	Sunflower	Leaf, Fruit	4	Real-world	–	–	RGB
Soybean Leaves Dataset	2021	6410	Soybean	Leaf	3	Real-world	Smartphones and Drones	500 × 500	RGB
Sweet Pumpkin Recognition Dataset	2024	7000[a]	Sweet pumpkin	Leaf	5	Laboratory	–	–	RGB

[a] Augmented.

Table 3
Access links for laboratory and real-world plant leaf disease datasets.

Dataset	Access link
BRACOL	https://data.mendeley.com/datasets/yy2k5y8mxg/1
RoCoLe	https://data.mendeley.com/datasets/c5yvn32dzg/2
Plant Pathology	https://www.kaggle.com/c/plant-pathology-2020-fgvc7/data
Rice leaf diseases	https://archive.ics.uci.edu/dataset/486/rice+leaf+diseases
PlantVillage	https://www.kaggle.com/datasets/abdallahalidev/plantvillage-dataset/code
Cassava Disease Leaf Dataset	https://www.kaggle.com/competitions/cassava-disease/overview
Citrus Dataset	https://data.mendeley.com/datasets/3f83gxmv57/2
DiaMOS Plant Dataset	https://zenodo.org/record/5557313
Taiwan tomato Leaves	https://data.mendeley.com/datasets/ngdgg79rzb/1
Rice Leaf Disease Image Samples	https://data.mendeley.com/datasets/fwcj7stb8r/1
Crop Pest and Disease Detection	https://data.mendeley.com/datasets/bwh3zbpkpv/1
Sugarcane Leaf Image Dataset	https://data.mendeley.com/datasets/9twjtv92vk/1
Cucumber Disease Recognition Dataset	https://data.mendeley.com/datasets/y6d3z6f8z9/1
MangoLeafBD Dataset	https://data.mendeley.com/datasets/hxsnvwty3r/1
Betel Leaf Image Dataset	https://data.mendeley.com/datasets/g7fpgj57wc/1
Sunflower Leaf Dataset	https://data.mendeley.com/datasets/b83hmrzth8/1
Soybean Leaves Dataset	https://data.mendeley.com/datasets/bycbh73438/1
Sweet Pumpkin Disease Recognition Dataset	https://data.mendeley.com/datasets/7397w9h5tb/1

4.1.8. Cassava disease leaf dataset

The cassava disease leaf dataset was gathered from various regions of Uganda, comprising 12,595 unlabeled and 9436 labeled images of cassava plant leaves (Mwebaze et al., 2019). Expert labeling was conducted by experts from the National Crops Resources Research Institute (NaCRRI). These images were classified into five categories: Healthy, Cassava Mosaic Disease (CMD), Cassava Brown Streak Disease (CBSD), Cassava Bacterial Blight (CBB), and Cassava Green Mite (CGM). They were captured using the same type of smartphone by different farmers from different real-world backgrounds over one year, and over different time frames.

4.1.9. Taiwan tomato leaves dataset

The dataset from Taiwan Tomato Leaves comprises 622 images of tomatoes, which are classified into six categories: five for various diseases and one for healthy tomatoes (Huang and Chang, 2020). These images were captured in various settings, including real-world environments and controlled laboratory conditions, showcasing a range

of backgrounds from complex to simple. Initially taken in varying sizes, the images were standardized to dimensions of 227 × 227. By employing data augmentation methods such as brightness adjustments, mirroring (horizontal and vertical), and rotation (90°, 180°, 270°), the dataset was expanded to include a total of 4976 images.

4.1.10. Rice leaf diseases dataset

The Rice Leaf Diseases dataset, collected from various rice fields in western Odisha (Sethy, 2020), comprises 5932 disease instances classified into four categories: Bacterial Blight, Blast, Brown Spot, and Tungro. Utilizing a Nikon DSLR-D5600 with an 18–55 mm lens, high-resolution field images were captured and standardized to 300 × 300 pixels to ensure consistency. Specifically, patches representing diseased regions were extracted from the original images. Furthermore, a portion of rice disease images was sourced from the Agricultural Pest and Insect Pests Picture Database (http://bcch.ahnw.gov.cn/Right.aspx).

4.1.11. Dataset for crop pest and disease detection

The dataset described in Mensah et al. (2023) is organized into two primary sections: raw images and augmented images. The raw dataset consists of 24,881 samples, while the augmented dataset includes 102,976 samples, distributed across four plant species with 22 distinct classes. All images have been de-identified and validated by expert plant virologists. The images were captured using a high-resolution camera, resulting in varied dimensions, including (400 × 400), (487 × 1080), (1080 × 518), (3024 × 4032), and (4032 × 3024). These images were taken under diverse conditions and backgrounds, such as white, dark, illuminated, and natural settings.

4.1.12. Sugarcane leaf image dataset

The Sugarcane Leaf Dataset contains 7134 high-resolution JPEG images of sugarcane leaves, sized at 768 × 1024 pixels with a resolution of 72 dpi (Thite et al., 2024). It is organized into 12 classes, encompassing 10 distinct disease types along with categories for healthy and dried leaves. This extensive collection was compiled through thorough field surveys and some lab settings, capturing different disease stages and angles for accurate visual identification. The purpose of the dataset is to cover a wide range of common sugarcane leaf diseases, providing researchers with a valuable tool for disease detection and agricultural analysis.

4.1.13. Cucumber disease recognition dataset

The dataset comprises 1280 original cucumber images gathered from real fields, which were subsequently augmented using techniques such as flipping, shearing, zooming, and rotation, leading to a total of 6400 images (Sultana et al., 2022). It encompasses eight distinct cucumber classes: Anthracnose, Bacterial Wilt, Belly Rot, Downy Mildew, Pythium Fruit Rot, Gummy Stem Blight, Fresh leaves, and Fresh cucumber. The classification of diseases was conducted in collaboration with an expert from an agricultural institute.

4.1.14. MangoLeafBD dataset

The dataset contains 4000 images of mango leaves, each with a resolution of 240 × 320 pixels, and saved in JPG format. Out of these, approximately 1800 images depict distinct leaves, while the rest were created through zooming and rotation as required (Ahmed et al., 2023). It encompasses seven diseases: Anthracnose, Bacterial Canker, Cutting Weevil, Die Back, Gall Midge, Powdery Mildew, and Sooty Mould, alongside a healthy category, resulting in eight classes in total. Each class comprises 500 images. The data collection involved capturing images of mango trees using a mobile phone camera.

4.1.15. Betel leaf image dataset

The dataset is structured into two primary folders: Original and Augmented, comprising a total of 3589 images (Rashid et al., 2024). Within the Original folder, there are 1000 images evenly spread across four leaf conditions, ensuring balanced representation. The Augmented folder extends the dataset to 2589 images, split into training (2388 images) and testing (201 images) subsets. This structure facilitates the training of ML and DL algorithms to classify and recognize various health states of betel leaves accurately.

4.1.16. Sunflower leaf dataset

The dataset consists of two types of sunflower image files: original and augmented. Each file contains images of three diseases—downy mildew, leaf scars, and gray mold—along with healthy sunflower leaves (Rajbongshi et al., 2022). 467 images were collected from a sunflower garden for the original file. Through augmentation, the dataset was expanded to 1668 images in the augmented file. These images, depicting disease-affected and healthy sunflowers, were gathered from the Bangladesh Agricultural Research Institute (BARI) demonstration farm in Gazipur with the assistance of domain experts.

4.1.17. Soybean leaves dataset

The dataset comprises three folders of soybean leaf images: Caterpillar, Diabrotica Speciosa, and Healthy, totaling 6410 images (3309 for Caterpillar, 2205 for Diabrotica Speciosa, and 896 for Healthy) (Mignoni, 2021). Captured in natural environments using smartphones and drones, the images were standardized to 500 × 500 pixels. They were also augmented through rotations of 45, 90, and 180 degrees. Researchers can apply their filters and preprocessing tools to this raw data.

4.1.18. Sweet pumpkin disease recognition dataset

The initial dataset consisted of 1000 Sweet Pumpkin images collected from the field (Mafi, 2024). To expand the dataset, various image processing techniques—including shifting, flipping, zooming, shearing, brightness enhancement, and rotation—were applied, increasing the total to 7000 augmented images. This comprehensive dataset includes five distinct Sweet Pumpkin conditions: Healthy Leaf, Downy Mildew Disease, Leaf Curl Disease, Mosaic Disease, and Red Beetle. These diverse conditions are valuable for developing machine vision-based techniques.

4.2. Dataset comparative analysis

In this section, a comparative analysis of various datasets is presented. Comparison is performed across multiple factors, including dataset collection techniques, the number of plants, categories of diseases, and technical dataset settings. Insights are offered about the methodologies used in collecting the datasets, the diversity of plant species covered, the range of disease categories addressed, and the technical settings implemented in each dataset. Fig. 6 presents an overview diagram illustrating popular datasets (inner circle) utilized in plant disease recognition, along with their corresponding deep learning models (outer circle), as discussed in this survey.

4.2.1. Dataset collection technique

DL frameworks heavily rely on datasets to facilitate direct learning by their underlying algorithms. These datasets are commonly classified based on the background setting of the images: controlled environments yield laboratory datasets, whereas real-world contexts produce in-field datasets. In this study, a comparison is conducted among eighteen datasets: nine in real-world environments, six in laboratory conditions, and three collected in both scenarios. Currently, there is increasing attention on real-world datasets to ensure applications align with practical needs, reflecting a trend among researchers.

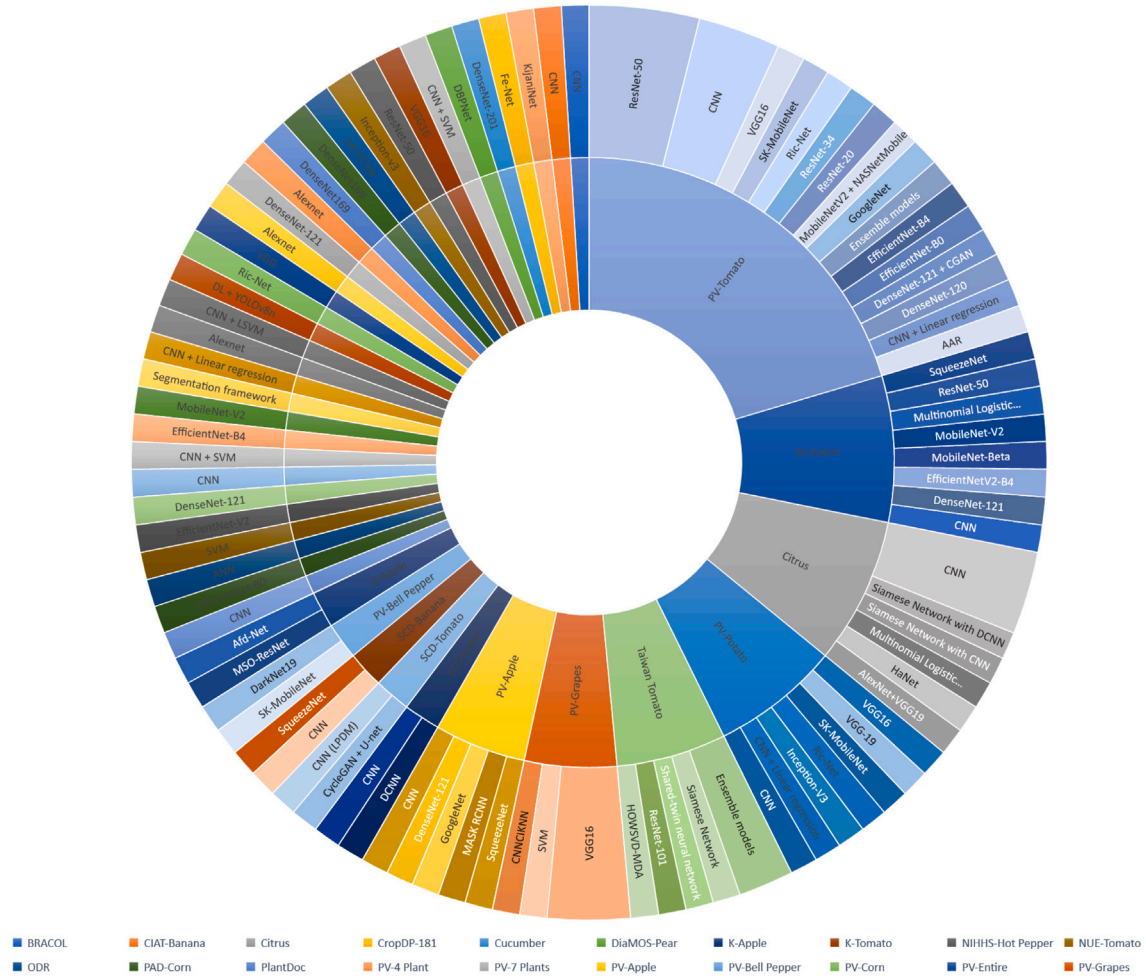


Fig. 6. An overview diagram showing popular datasets (inner circle) used in plant disease recognition and their respective deep learning models (outer circle) in this survey.

Another critical factor influencing performance is the quality of the capturing device, which includes factors such as pixel size, zoom capability, focusing method, and additional functionalities. Images taken in a controlled laboratory environment are often aimed at achieving a uniform or neutral background, precise lighting, and sharp focus on the foliage. In contrast, capturing images in real-world settings involves using various photography tools and dealing with environmental variables like wind, varying lighting conditions, and potential obstructions from adjacent plant foliage. Images from real-world environments require attention to diverse angles, surface textures, lighting variations, and noise levels during the capture process. This detailed examination offers valuable insights into the diverse attributes and challenges associated with datasets used in DL applications.

4.2.2. Number of plants and categories of diseases

Variations are observed in the number of plants and disease classes encompassed by plant disease datasets. The largest dataset, PlantVillage, comprises 54,390 samples across 14 different plants and 38 classes (Hughes et al., 2015). In contrast, many other datasets focus on a single plant, concentrating on a higher number of disease classes tailored to researchers' specific objectives. For instance, when considering the tomato in the PlantVillage dataset, includes nine disease classes, while the Taiwan Tomato Leaves dataset covers only five. As a result, the number of diseases for the same plant varies. Furthermore, the Taiwan Tomato Leaves dataset contains fewer samples due to various factors that led to limitations in collecting a larger number of samples and disease classes (Huang and Chang, 2020). Considerations during the dataset collection process include factors such as the availability of disease classes at the location, constraints in manpower, technical support, and limitations in research funding.

4.2.3. Technical dataset setting

In DL experiments, a large dataset is crucial for achieving high-efficiency performance. However, obtaining such a dataset poses various challenges due to factors such as the requirement for extensive manpower, a significant time frame and a substantial budget. The data collection process's important features are highlighted in Table 2, showcasing variations among datasets. Different capturing devices were employed in the data collection activities, depending on the budget or objectives of the research study. To address the challenge of a small number of data, some researchers have artificially increased the total sample size using data augmentation methods (Huang and Chang, 2020). Crowd-sourcing plant leaves from many farmers is another effective approach to collecting a large dataset. For instance, one research project engaged 200 farmers in capturing images using mobile phones, resulting in a collection of 22,031 images, demonstrating a successful strategy for amassing a large dataset (Mwebaze et al., 2019). Table 2 provides further technical details for comparing dataset collection processes.

4.3. Ideal dataset for deep learning

Researchers are currently focusing on plant disease classification work that requires processing in real-world environments to aid farmers and improve product quality. As a result, there is a necessity to employ

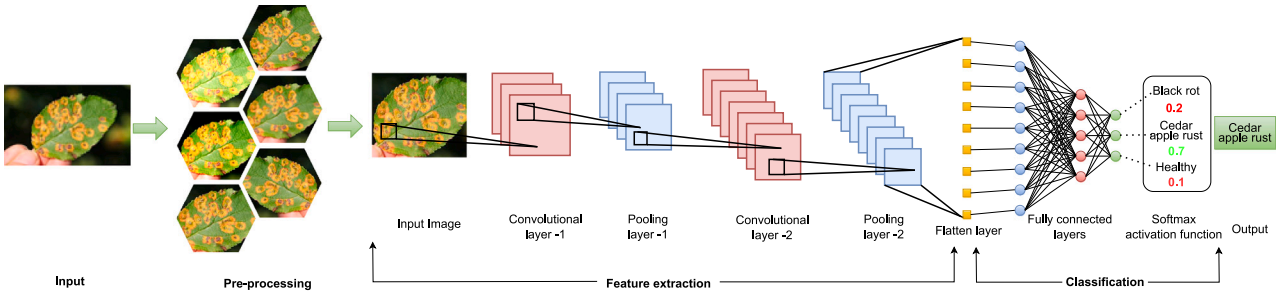


Fig. 7. The general pipeline outlining the fundamental steps involved in plant leaf disease recognition: preprocessing, feature extraction and classification.

field acquisition datasets (Singh et al., 2022b; Zhao et al., 2022; Algaashani et al., 2022; Zhao et al., 2021; Yadav et al., 2022; Paymode and Malode, 2022). However, several limitations are associated with these datasets, including issues with dataset size, imbalanced classes, and incompleteness (Fenu and Malloci, 2021; Mwebaze et al., 2019; Huang and Chang, 2020). Furthermore, it is recommended that ideal datasets consist of a diverse range of plant leaf diseases and include leaves collected at various stages of disease growth (Paymode and Malode, 2022). The timing of photography is crucial as it directs more attention to the structural features, including the color, size, and shape of the diseased leaves. Advanced sensors with high-resolution devices can be employed during image capture, considering various illumination, angles, surfaces, and noise conditions in the real-world environment.

The significant problem posed by imbalanced data in the DL framework necessitates solutions in future datasets to prevent the creation of biases among classes. Addressing this issue can be achieved through the implementation of an augmentation process in the dataset (Hasan et al., 2022; Kavala and Pothuraju, 2022; Kong et al., 2022). This technique aimed to increase the amount of data and improve the learning efficiency of the DL framework (Parraga-Alava et al., 2019; Thapa et al., 2020; Rauf et al., 2019; Fenu and Malloci, 2021). However, it is better to collect real datasets that are balanced rather than relying solely on augmentation. Real datasets ensure that the data's inherent characteristics and variations are accurately captured. Additionally, balanced real datasets help prevent the model from learning artificial patterns that may arise from augmentation techniques.

5. Preprocessing, feature extraction and classification

In this section, we provide a brief overview of preprocessing, feature extraction, and classification for DL-based plant disease recognition, focusing on the steps of data preprocessing, deep feature extraction, and plant disease classification. The general pipeline of plant disease recognition researchers comprised of these steps is illustrated in Fig. 7. Preprocessing techniques help improve the quality of input images, ensuring better performance of subsequent stages. Deep feature extraction involves extracting meaningful representations from the input data, capturing intricate patterns and characteristics essential for disease identification. Finally, plant disease classification assigns the input images to specific disease categories based on the extracted features, enabling automated diagnosis and timely intervention. A brief review of existing preprocessing, deep feature extraction and classification techniques used in plant leaf disease recognition is provided in the following subsections.

5.1. Pre-processing

In DL models, large datasets with high-quality images are required for effective learning. However, practical challenges are posed to researchers when collecting sufficiently large datasets. Weaknesses may be exhibited by newly collected datasets, such as an insufficient number of samples in certain classes, noisy images, and images with background objects. Consequently, preprocessing of the dataset is essential before model training to enhance data quality and optimize model performance.

5.1.1. Resizing and cropping

§Nodes in neural networks learn features from image pixels. To shorten model training time, resizing large pixel images is recommended, as noted in Saponara and Elhanashi (2022), though maintaining standard image sizes is crucial for accuracy, as highlighted in Luke et al. (2019). The image size can be varied due to factors like capturing devices, zooming, resolutions, distances, angles, and surfaces during data acquisition (Kong et al., 2022; Krishnan et al., 2022). In DL networks, some popularly used CNN architectures' optimized image sizes include 227 × 227 for AlexNet, 224 × 224 for DenseNet, ResNet, and VGG, and 299 × 299 for Inception. Therefore, resizing images according to the specifications of the training model is deemed essential (Boulent et al., 2019).

A range of elements, including diseased regions, non-diseased sections, and background details, are included in the images taken (Fenu and Malloci, 2021; Gu et al., 2021). The model must focus solely on the diseased spots to effectively detect diseases (Gardie et al., 2022). Cropping facilitates the isolation of relevant areas within the image, allowing the model to concentrate on the features crucial for disease identification. By focusing solely on the diseased regions, the model can eliminate irrelevant background noise, leading to more precise and reliable diagnosis outcomes. This targeted approach not only improves the efficiency of disease recognition but also reduces computational load and resource requirements (Gu et al., 2021).

5.1.2. Noise handling

The process of classification heavily relies on obtaining high-quality images from the dataset, yet images often contain noise, posing a significant challenge in effectively training the model (Shantkumari and Uma, 2022). Hence, it becomes crucial during pre-processing to eliminate noise from the images. This not only enhances image quality but also improves the efficacy of key feature extraction (Khan et al., 2022). To achieve this, commonly utilized filters such as the median and average filters are employed to eliminate gray or salt and pepper noise from the images (Krishnan et al., 2022; Narayanan et al., 2022). Moreover, other filters like the Gaussian filter are used to blur images and eliminate noise (Ashok et al., 2020), while the Wiener filter is specifically employed to reduce noise clusters within the image (Archana and Sahayadhas, 2018). By systematically applying these filters during pre-processing, the noise in images is effectively reduced, thereby enhancing the overall quality of the dataset and facilitating more accurate classification results.

5.1.3. Augmentation

Dealing with insufficient or imbalanced data, DL frameworks present a significant challenge for researchers working with real-world or laboratory datasets. Effective model training can be hindered by

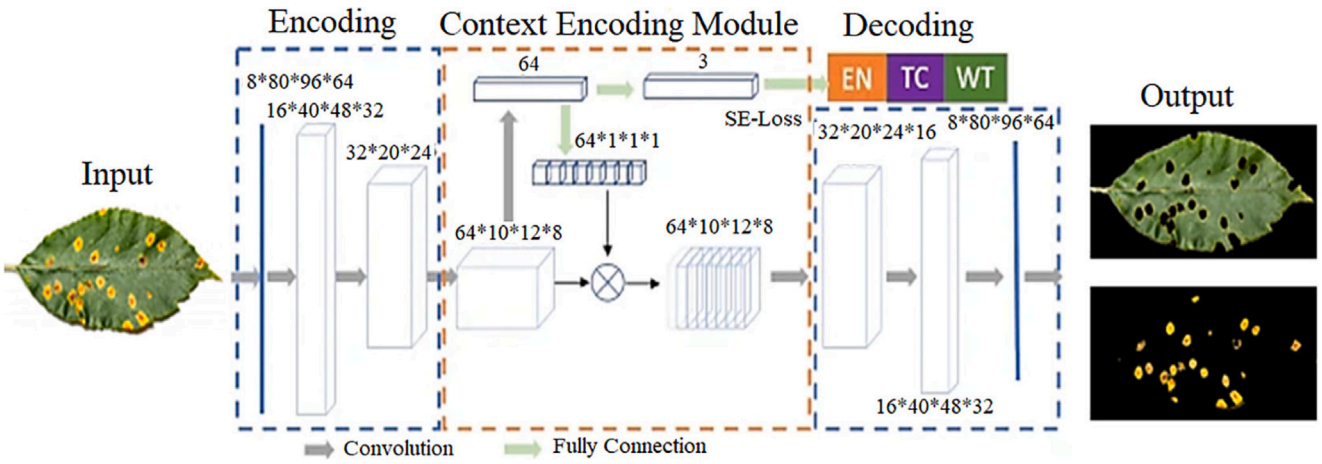


Fig. 8. A typical example of dedicated background removing approach: CANet neural network-based disease detection and ROI segmentation. *Source:* As extracted from Shoaib et al. (2022).

this scarcity of data within specific classes. To address this challenge, researchers employ various data augmentation techniques aimed at augmenting the dataset and balancing class distributions. Given that DL frameworks typically require a substantial amount of data for improved accuracy, these augmentation strategies play a crucial role in mitigating the effects of data scarcity and imbalance to a certain extent.

A diverse range of data augmentation techniques is employed to enhance the robustness and generalization capabilities of deep learning (DL) models. These techniques encompass both traditional augmentation methods and advanced strategies. Traditional data augmentation approaches are broadly classified into position-based and color-based techniques. Position-based methods, such as mirroring, clipping, and rotation, along with color-based techniques involving adjustments to brightness, contrast, and saturation, have been effectively applied in prior studies (Kavala and Pothuraju, 2022; Kong et al., 2022; Gardie et al., 2022; Esgario et al., 2022).

Additionally, methods like CutMix and MixUp are used to further augment the dataset. Sample images are combined using linear interpolation in the MixUp algorithm, creating new composite images. Conversely, CutMix involves removing pixel regions from one image and patching them onto another image, thereby generating new images and corresponding labels (Yadav et al., 2022; Kong et al., 2022). Second, advanced techniques such as Generative Adversarial Networks (GANs) and Neural Style Transfer (NST) are now being integrated into the data augmentation process for deep neural networks (Pandian et al., 2022; Nazki et al., 2019; Abbas et al., 2021; Prasad et al., 2022).

5.1.4. Background removing

The presence of background objects in images was found to affect both model training and the final classification tasks adversely. Consequently, researchers emphasized the importance of removing these background objects to optimize DL model training and improve the classification process by effectively directing the model's attention to the region of interest. Barbedo (2018) proposed a method to evaluate the impact of background masking. Backgrounds were manually removed from the images, and datasets with background-masked images achieved higher accuracy compared to those with backgrounds. However, some classes exhibited higher accuracy even with background images present. Manso et al. (2019) conducted a study on image segmentation, utilizing the k-means algorithm, Otsu method, and iterative thresholding techniques in the YCgCr color space. Kc et al. (2021) focused on segmentation tasks, employing edge and morphological-based techniques to effectively remove unwanted elements such as plants, soil, rocks, and human or animal body parts. This approach reduces noise in the training dataset and lowers resource costs, enhancing classification accuracy.

Recent research has witnessed the integration of background removal processes into DL models like U-net, Deep LabV3, Mask R-CNN, and PSPNet, demonstrating significant improvements over traditional methods. Sunil et al. (2021) utilized a U-net approach for background removal on a real-life cardamom plant disease dataset, incorporating object cropping during the segmentation phase. Wu et al. (2023) employed the DBPNet model for segmenting a dataset of pear leaves, achieving superior results compared to other models such as U-net, DeepLabV3, MANet, and FPN. Additionally, the KijaniNet model utilized segmentation, with experiments conducted using both background-included and background-removed images for classification, revealing that segmentation significantly increased accuracy (Ngugi et al., 2023).

Numerous background-removal techniques have been proposed to enhance plant leaf disease recognition. The CANet neural network is a notable method, leveraging background removal for improved disease detection (Shoaib et al., 2022). Its architecture, shown in Fig. 8, details the input, hidden, and output layers, emphasizing its role in region of interest (RoI) segmentation. This approach represents a significant advancement in plant disease recognition

5.1.5. Normalization

The variations in intensity values and dimensions in images are standardized through a process called normalization, which reduces pixel values to a smaller range. This normalization is crucial in DL training to optimize the model's efficiency. By dividing pixel values by the maximum value (usually 255), a standardized range of 0 to 1 is ensured through normalization, aiding faster learning and improved model performance in object extraction from images (Singh et al., 2022a; Koo and Cha, 2017). Eq. (1) expresses the Z-score normalization method.

$$Zi = \frac{x(i) - \bar{x}}{\delta(x)} \tag{1}$$

The sample values, channel means, and channel standard deviation, denoted as $x(i)$, $\bar{x}$, and δx, respectively, are used in subtracting the sample value from the channel mean and dividing by the standard deviation value for the channel pixels (Chao et al., 2020). An alternative normalization technique known as linear normalization, also referred to as "Max-Min" normalization, is denoted as Eq. (2).

$$Fnorm = \frac{F - F(min)}{F(max) - F(min)} \tag{2}$$

where, $F(min)$, $F(max)$, and F were denoted minimum pixel intensity value, maximum pixel intensity value, and pixel intensity value (Kaur et al., 2022).

5.2. Deep feature extraction

The characteristics of an object, such as its components and patterns, are essential to accurately classifying images. DL models have shown exceptional performance in tasks such as image classification because they can automatically extract intricate, high-level abstract features from extensive datasets (Najafabadi et al., 2015). Similarly, in the domain of plant leaf disease recognition, feature extraction is essential for accurately recognizing diseases. Manual feature extraction was used in traditional ML techniques, focusing on aspects such as color and shape (Mutlag et al., 2020). These methods relied heavily on human expertise to identify and select relevant features from the images, which were then fed into classifiers to distinguish between different categories. Although effective to some extent, this approach had limitations, such as being time consuming and less adaptable to new or complex patterns.

In contrast, the feature extraction process is automated by DL models (Toda and Okura, 2019). DL models, particularly CNNs, can automatically learn and extract relevant features from raw images that eliminates the need for manual intervention. This automation is especially beneficial in the field of plant leaf disease recognition, where the complexity and variability of disease symptoms can make manual feature extraction challenging. By leveraging the capability of DL algorithms, models can identify patterns in leaf images that may be indicative of specific diseases. This not only improves the performance of disease classification, but also accelerates the diagnostic process.

5.3. Plant disease classification

In the domain of DL models, the classification process plays an important role in determining the overall performance of the model. Although DL models have the advantage of operating in an end-to-end manner, learning features and performing classification simultaneously—traditional methods take a different approach. Traditional ML techniques typically separate feature extraction and classification into distinct steps. They often rely on various classifiers, such as SVM, k-nearest neighbors (KNN), logistic regression, and random forest, to perform the classification task. These classifiers are used extensively in plant disease classification, where they analyze manually extracted features like color, texture, and shape to identify disease patterns. However, despite their widespread use, traditional classification methods generally yield lower accuracy compared to DL models.

Accuracy is one of the most important parameters in the classification process (Singh et al., 2022a). The inclusion of a softmax function in the classification layer of any DL model is highly beneficial for transforming scores into standardized probabilities. Additionally, the use of the cross-entropy loss function significantly enhances the performance metrics of the model. While DL models often function seamlessly as end-to-end methodologies, supplementary classifiers like SVM are sometimes integrated with CNN architectures. This additional classifier, known for its proficient accuracy even with limited data, serves as a benchmark for evaluating the performance of the DL model (Gui et al., 2021). For example, a CNN model paired with an SVM classifier effectively addressed the challenge of declining accuracy caused by imbalanced datasets (Su et al., 2019). Despite this, DL models automatically learn complex patterns and relationships within the data, leading to more precise and reliable recognition of plant diseases from leaf images.

6. Deep learning-based plant disease recognition

Significant advances in plant disease recognition have been led by the advent of DL techniques, marking a breakthrough in research. In addition, the transition from traditional methods to DL has been made feasible by the availability of datasets, enhanced computational capabilities, and software that supports complex DL architectures with reduced complexity. Fig. 9 presents a comparative analysis of the deep learning architectures employed in the reviewed studies, highlighting the diversity of models applied in the surveyed literature. In this section, we compare various DL-based plant leaf disease recognition methods, analyzing their performance in architecture and dataset-specific contexts.

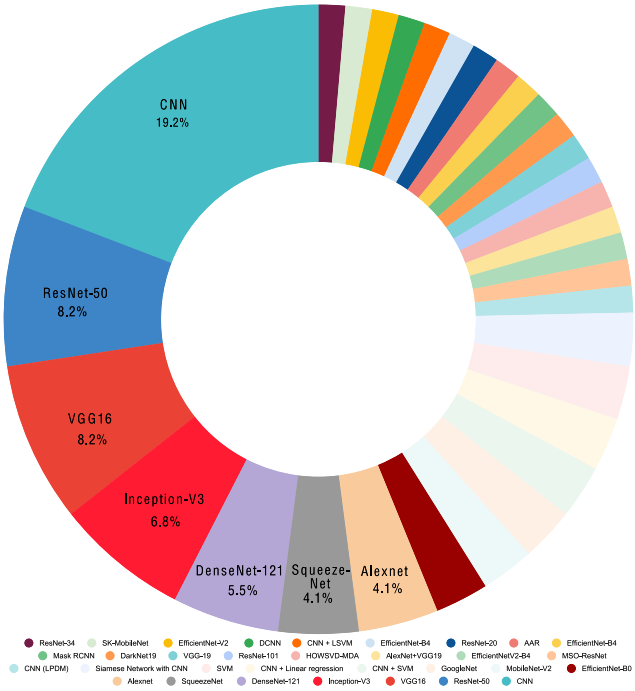


Fig. 9. Comparison of popular models in this survey for plant disease recognition.

6.1. Deep architecture-specific analysis

The rapid evolution of DL has led to the development of various architectures, each designed to optimize performance for specific tasks. Among these, AlexNet, VGGNet, DenseNet, GoogLeNet, ResNet, MobileNet, SqueezeNet, and EfficientNet stand out in the plant leaf disease recognition domain, as each architecture brings distinct advantages. This subsection provides an in-depth, architecture-specific analysis of these prominent DL models and highlights their prominence in plant leaf disease recognition.

6.1.1. AlexNet

AlexNet, a CNN-based model proposed in 2012, is considered a landmark model in the history of DL (Krizhevsky et al., 2012). It was designed to outperform its contemporaries, utilizing 60 million parameters and 650,000 neurons during training. The architecture is composed of eight layers, with five convolutional layers and three fully connected layers, all incorporating the Rectified Linear Unit (ReLU) activation function. The effectiveness of the ReLU function is evident, as a 25% training error was achieved on the CIFAR-10 dataset in the fourth convolutional layer, significantly surpassing a similar network with tanh neurons by a factor of 6. The AlexNet model processes input images sized at $224 \times 224 \times 3$ using $11 \times 11 \times 3$ convolutional layers with a stride of 4 pixels, containing 96 kernels. The following layers consist of 256 kernels, then 384 kernels twice, and ending with 256 kernels. By incorporating local response normalization and max pooling layers, performance is boosted, while training efficiency is enhanced through multi-GPU utilization. To address the issue of overfitting, the fully connected layers employ a regularization technique called dropout, which has proven highly effective. In evaluations, AlexNet achieved a

top-5 error rate of 15.3%, notably lower than the second-place error rate of 26.2%. Modifications were made to the AlexNet architecture by focusing on alterations in the initial four layers, crucial within the CNN framework, aimed at reducing memory requirements and enabling efficient execution of the model on the Android platform (Chen et al., 2022). This influential model significantly impacted the development of DL in many domains (Krizhevsky et al., 2017).

In the domain of plant leaf disease recognition, the AlexNet model was used in Singh et al. (2022b) to classify maize plant diseases, achieving an impressive accuracy of 98.75% compared to VGG-16, ANN, and SVM models. Alruwaili et al. (2019) used AlexNet for the classification of plant leaf disease, applying it to the PlantVillage dataset, resulting in an outstanding overall accuracy of 99.11%. Similarly, the AlexNet model was employed in Jadhav et al. (2021) for soybean leaf disease classification, achieving an accuracy of 98.75% in their study, surpassing the performance of GoogleNet.

6.1.2. VGGNet

The VGG deep network, developed in 2014, is another significant CNN-based architecture that determined optimal configurations through experiments involving varying layer numbers (Simonyan and Zisserman, 2014). The most effective configuration consisted of 16 and 19 layers, known as VGG-16 and VGG-19. This model operates on RGB input images sized $224 \times 224 \times 3$ and it utilizes 3×3 convolutional filters across all layers with a stride of 1 pixel, alongside 2×2 max-pooling layers with a stride of two. The inclusion of the ReLU activation function is crucial to the VGG architecture, introducing nonlinearity that enhances its classification capabilities and significantly boosts computational efficiency compared to earlier DL frameworks. In particular, the VGG-16 achieved an impressive top-5 test error of 7.0%, establishing itself as the leading single-net model.

In Kavala and Pothuraju (2022), VGG-16 and VGG-19 are used for the plant leaf disease recognition task, with VGG-16 demonstrating superior performance by achieving the highest accuracy of 98.75% in classifying grape leaf diseases. Nishad et al. (2022) proposed an approach using K-means clustering segmentation, various data augmentation techniques, and the VGG-16 model to detect and classify potato leaf diseases. Their approach achieved a high accuracy of 97%, outperforming the VGG-19 model on the Potato Leaf Disease dataset. In Paymode and Malode (2022), the VGG-19 model is used to recognize plant leaf diseases. They focused on the classification of grape and tomato leaf diseases, achieving superior accuracy using the transfer learning method with VGG16: 98.40% for grapes and 95.71% for tomatoes, surpassing the performance of Inception-v1, ResNet-V2, AlexNet, and DCNN DL models.

6.1.3. DenseNet

Huang et al. (2017) introduced the DenseNet in 2017 with a distinctive architecture in which every layer is directly connected to all other layers in a forward flow manner. This arrangement yields N connections for the N layers, resulting in a total of $N(N+1)/2$ direct connections within the network. DenseNets receive feature maps as input from preceding layers, establishing these direct connections. The composite functions utilized in DenseNets encompass ReLU activation, Batch Normalization, pooling, and convolution. During the pooling stage, a batch normalization function is succeeded by a 1×1 convolutional layer and then a 2×2 average pooling layer. The model is evaluated on four rigorous object recognition benchmarks, CIFAR-10, CIFAR-100, SVHN, and ImageNet. The DenseNet family includes variants such as DenseNet-121, DenseNet-169, DenseNet-201, and DenseNet-264, each differing in the number of layers within the dense blocks.

Automated plant disease recognition studies commonly used DenseNet, a particularly deep framework known for its substantial number of layers. Exceptional results, including an accuracy of 99.68%, were achieved by a pre-trained DenseNet-120 network, surpassing various models such as ResNet-101, ResNet-50, ResNet-30, ResNet-18, SqueezeNet, and VGGNet (Aquil and Ishak, 2021). To further enhance performance, in Nandhini and Ashokkumar (2022), the hyperparameters of the DenseNet-121 architecture are optimized using the Mutation-based Henry Gas Solubility Optimization (MHGSO) algorithm.

6.1.4. GoogLeNet

The GoogLeNet architecture, introduced in 2014, was designed with increased depth and width and won the image classification task in the ILSVRC-2014 competition. Operated on RGB color channels with a size of 224×224 and employing mean subtraction across 22 layers, GoogLeNet utilizes small convolutions of sizes 1×1, 3×3, and 5×5. Its efficient design allows for effective operation even with limited computational resources. The architecture includes 3×3 max-pooling layers and 1×1, 3×3, and 5×5 convolutional layers for feature extraction, with integrated dimension reduction modules to lower computational costs. The concatenated 1×1, 3×3, and 5×5 convolutional layers, along with the inception module, produce a single vector output. Noteworthy modifications in the inception module involve adding a 1×1 convolutional layer before another layer and employing a max-pooling layer in parallel (Szegedy et al., 2015).

Maeda-Gutiérrez et al. (2020) applied transfer learning with the GoogLeNet model to identify tomato leaf diseases, achieving an accuracy of 99.39%. The fine-tuning step specifically targeted the final three layers, comprising a fully connected layer, a SoftMax layer, and a classification output layer. With a total of 7 million parameters spread over 22 layers, this approach demonstrates effective plant disease detection capabilities. Baranwal et al. (2019) proposed a model using the GoogLeNet architecture, with the best accuracy achieved by the model being 98.42% on the PlantVillage dataset for the apple plant.

6.1.5. ResNet

Introduced in 2015, ResNet was designed to mitigate accuracy degradation in DL models due to vanishing or exploding gradients as the number of layers increases (He et al., 2016). The shortcut connection technique introduced in ResNet allows the identities x to bypass layers, using the new input as the layer output $F(x)$ and adding the identity value x to the new input as $F(x) + x$. ResNet models, with versions of 18, 34, 50, 101, and 152 layers, effectively prevent the vanishing or exploding gradient issue by incorporating skip connections.

Yilma et al. (2020) employed the ResNet-50 architecture in conjunction with a two-pathway Encoder GAN data generation network to classify tomato plant leaf diseases, attaining the highest accuracy of 98.74% compared to AlexNet, GoogLeNet, and VGG-19 in terms of accuracy. Building on this, Zhao et al. (2021) enhanced the ResNet-50 framework by integrating the Squeeze-and-Excitation (SE) attention, effectively extracting complex features from various tomato leaf diseases. In Gu et al. (2021), a pre-trained ResNet-50 model was used on cropped disease areas, followed by employing a KNN algorithm to extract deep features for pests and 19 types of hot-pepper diseases.

6.1.6. MobileNet

The MobileNet architectures, introduced by Google, are specifically designed for optimal performance on resource-constrained devices such as mobile devices. MobileNet-v1 is composed of two layers: a depthwise convolution layer for lightweight filtering, followed by a pointwise convolution layer to generate new features. MobileNet-v2, an upgraded version, features two blocks, each comprising three layers: a 1×1 convolution layer with ReLU6, a depthwise convolution layer, and a final 1×1 convolution layer devoid of nonlinearity. Its inverted residual structure integrates bottleneck layers and shortcut connections, resulting in enhanced performance compared to models featuring shortcut connections at expanded layers. During training, 3.4 million parameters are employed by the model, and the linear

bottlenecks optimize performance by efficiently reducing information in low-dimensional space (Howard et al., 2017; Sandler et al., 2018).

In Chen et al. (2021), MobileNet-v2 with the SE attention block was used to detect diseases in rice plants, achieving an average accuracy of 99.78% on the Plant Village dataset and 99.33% on a real-life rice disease dataset, with results compared to Inception-v3, VGGNet, DenseNet, NASNetMobile, and MobileNet-v2. Another lightweight strategy known as selective kernel MobileNet (SK-MobileNet) incorporates the SK block, which integrates Split, Fuse, and Select operations, significantly enhancing model performance (Liu et al., 2023). It achieved 99.28% accuracy for identifying three types of plants—bell pepper, potato, and tomato—from the PlantVillage dataset. In Younis et al. (2021), additional robust optimization techniques and MobileNet-v2 were employed. The model was initially trained on the ImageNet dataset and fine-tuned on 38 classes of 14 different plants from the PlantVillage dataset, achieving an average accuracy of 98.89%. MobileNet-Beta, an extension of MobileNet-v2 with a classification activation map for improved plant disease detection introduced in Chen et al. (2020), demonstrated its effectiveness by achieving accuracies of 99.85% on the PlantVillage dataset and 99.11% on a proprietary dataset.

6.1.7. SqueezeNet

SqueezeNet presents a distinctive structure with 8 fire modules enclosed by two convolutional layers, where each fire module contains a squeeze convolutional layer using a 1×1 filter, followed by an expanded layer integrating 1×1 and 3×3 convolutional filters, with the filter count progressively increasing across the network (Iandola et al., 2016). Max-pooling with a stride of 2 is followed by a convolutional layer and 2 fire modules, utilizing ReLU as the activation function and integrating dropout after the final fire module.

Aravind et al. (2020) used a SqueezeNet model on the Plant Village dataset, achieving a top classification accuracy of 98.49% using the entire dataset. In another study, Bhuiyan et al. (2023) used SqueezeNet for classifying banana leaf diseases. A real-world dataset was used and augmentation techniques were implemented to address the imbalance problem, resulting in the model achieving an accuracy of 96.25%, which was then compared to the performance of EfficientNet-B0, MobileNet-v3, ResNet-101, ResNet-50, InceptionNet-v3, and VGG-16.

6.1.8. EfficientNet

EfficientNet, known for its efficient scaling across depth, width, and resolution, achieved an impressive top accuracy of 84.33% on the ImageNet classification task using 66 million parameters (Tan and Le, 2019). A significant advancement has been made in the compound scaling technique, which uniformly adjusts the model dimensions to achieve an optimal balance, resulting in higher accuracy without a disproportionate increase in computational requirements. EfficientNet includes eight models, denoted from B0 to B7, distinguished by their novel strategy for systematically and proportionally adjusting the model's depth, width, and resolution. With parameter counts ranging from 4 million to 64 million across B0 to B7, EfficientNet B0 was identified as an exceptionally effective model, particularly suited for tasks requiring fewer parameters.

Rajeena PP et al. (2023) used an EfficientNet-B0 model to detect corn leaf diseases, achieving an accuracy of 98.85%, which was notably superior when benchmarked against Inception, VGG-16, ResNet-50, ResNet-101, ResNet-152, and DenseNet-121. Hassan et al. (2021) tested a range of learning rates (0.01 to 0.0001) to determine their efficiency in training several models, including EfficientNet-B0. Using the Plant Village dataset, the EfficientNet-B0 model achieved an accuracy of 99.56%. This model was compared with Inception-v3, InceptionResNet-v2 and MobileNet-v2, revealing that both MobileNet-v2 and EfficientNet-B0 required less training time while maintaining high accuracy. In Atila et al. (2021), all variations of EfficientNet models were compared using the PlantVillage dataset, covering various crops, and included both the original and augmented datasets in the experiment. In the original dataset, the EfficientNet-B5 model achieved an accuracy of 99.91%, while in the augmented dataset, the EfficientNet-B4 model achieved an impressive accuracy of 99.97%. EffiMob-Net, a hybrid DL model combining EfficientNet-B3 and MobileNet, was introduced in Ullah et al. (2023) to accurately detect tomato plant diseases from leaf images. The EffiMob-Net demonstrated significant improvements in recognizing diseases from plant leaves.

6.2. Dataset-specific analysis

Agriculture's significance drives governments and non-governmental organizations to prioritize developing benchmark datasets for plant disease recognition, supporting effective automated methods (Steensland and Zeigler, 2021). There are several benchmark datasets available for plant leaf disease recognition; notable among them is PlantVillage, which includes 38 classes across 14 different plant species. Automated plant leaf disease recognition techniques are evaluated based on plant types, with model performance assessed across commonly used datasets. Therefore, in this subsection, various plant leaf disease recognition methods for different plant types are analyzed in a dataset-specific manner.

6.2.1. Tomato - PlantVillage

In the literature, 10 common classes of tomato leaf diseases are analyzed, reflecting the global importance of tomatoes and increasing researcher emphasis on achieving improved disease recognition. Chowdhury et al. (2021) utilized the tomato sub-dataset from the PlantVillage dataset in their research, employing transfer learning architectures to conduct binary classification, six-class classification, and ten-class classification, with the EfficientNet-B4 model demonstrating exceptional performance by achieving an accuracy of 99.89%. In Tan et al. (2021), the Score-CAM visualization technique was applied to enhance the classification of tomato leaves, with the ResNet-34 model showcasing an impressive accuracy of 99.7%. It was compared with traditional ML classifiers such as KNN, SVM, and Random Forest (RF), as well as DL models such as AlexNet, VGG-16, EfficientNet-B0, and MobileNetV2. In another work, the Convolutional Block Attention Module (CBAM) approach, along with a lightweight concept, was incorporated into ResNet-20 architectures, achieving an impressive accuracy of 99.69% in tomato leaf disease recognition (Bhujel et al., 2022).

An F1-score of 99.92% was achieved in Aquil and Ishak (2021) using ResNet-101, which included both from-scratch and pre-trained CNNs, for the recognition of tomato leaf disease. The results are compared against various DL models, such as VGG-16, VGG-19, Squeezenet, ResNet-18, ResNet-34, ResNet-50, and Densenet-120. In Maeda-Gutiérrez et al. (2020), pre-trained deep-learning models such as AlexNet, GoogleNet, Inception V3, ResNet 18, and ResNet 50 were fine-tuned to recognize diseases of tomato plant leaves, with the GoogleNet architecture achieving the highest accuracy of 99.39%. Harakannanavar et al. (2022) used various descriptors, including the Discrete Wavelet Transform (DWT), Gray Level Co-occurrence Matrix (GLCM), and Principal Component Analysis (PCA), for feature extraction of tomato leaves, and employed a CNN framework for classification, achieving an accuracy of 99.09%.

Yilma et al. (2020) proposed an innovative method for recognizing tomato diseases using an Attention-Augmented Residual (AAR) network combined with a Conditional Variational GAN (CVGAN) for sample reconstruction and augmentation. This approach effectively addressed the issues of training sample scarcity and class imbalance, achieving an accuracy of 98.91%. In Abawatew et al. (2021), the network integrates a pre-activated residual block for coarse-level feature learning with an attention block to capture significant features and global associations. Zhao et al. (2021) incorporated attention mechanisms into deep CNN models for tomato image augmentation, using residual and attention modules to recalibrate channel-wise feature responses adaptively. The

Table 4
A comparative analysis of different DL approaches was conducted on tomato plant leaf diseases using data from the PlantVillage dataset.

Authors	Architecture	Highlights	Description	Result
Chowdhury et al. (2021)	EfficientNet-B4	Segmentation: Modified U-net, Visualization: Score-CAM, Classification: EfficientNet-B4	Transfer learning was used to enhance accuracy. EfficientNet-B4 had its final layer fine-tuned and used Score-CAM for visualization. The Modified U-Net Model excelled in segmentation tasks.	Proposed model: 99.89% CNN: 91% LeNet: 94%
Tan et al. (2021)	ResNet-34	Comparison with ML classifier, color + GLCM feature extraction method given best result	ML and DL methods were compared in the classification process. The best result was yielded by the feature extraction processes using the color + GLCM method. The best accuracy was shown by the ResNet34 model.	Proposed model: 99.7% AlexNet: 92.7% VGG-16: 98.9% KNN: 82.1% SVM: 91% EfficientNet-B0: 98.9%
Bhujel et al. (2022)	ResNet-20	Lightweight model with used CBAM	The ResNet-20 model is designed as a lightweight model, with only 1.440813 million parameters utilized across 20 layers. Additionally, the CBAM is incorporated for identifying key feature regions.	Proposed model: 99.69% AlexNet: 98.93% GoogleNet: 99.39% Inception-v3: 98.65% ResNet-18: 99.06%
Maeda-Gutiérrez et al. (2020)	GoogleNet	Comparing the performance of the CNN model using transfer learning	The research was focused on comparing the performance of the transfer learning approach in DL models. Better accuracy was demonstrated by utilizing 22 layers with 7 million parameters compared to other CNN models.	Proposed model: 99.39% AlexNet: 98.93% Inception-v3: 98.65% ResNet-18: 99.06%
Abawatew et al. (2021)	AAR	CVGAN, Attention heat map and t-SNE visualization	The CVGAN method was utilized by the AAR network for sample reconstruction. Data imbalance issues were addressed by the ARR network. The Attention heat map and t-SNE visualization process were employed by the AAR network.	Proposed model: 98.91% ResNet-50: 98.74% Attention CNN: 98.00%
Yilma et al. (2020)	ResNet-50	Two pathway encoder GAN	Challenges were posed by the imbalanced data and the scarcity of data in DL frameworks, thus necessitating the employment of the two-pathway Encoder GAN method to overcome these issues and achieve performance.	Proposed model: 98.74% AlexNet: 87.97% GoogleNet: 96.98% VGG-19: 98.74%
Zhao et al. (2021)	ResNet-50	Attention module, The Squeeze-and-Excitation Networks	The SE-ResNet-50 was employed to extract complex features. The experiment included the classification of both tomato leaves and grape leaves. Superior performance in grape leaf disease classifications compared to tomato leaf disease classifications was demonstrated by the model.	Proposed model: 96.81% ResNet-50: 92.56% GoogleNet: 87.27% Xception: 88.16% VGG-16: 90.42%
Ullah et al. (2023)	ResNet-50	GAN based ResNet-50-v2 with Generator and Discriminator	GAN is used for data feature selection and as a data augmentation technique to prevent overfitting.	Proposed model: 99.75% EfficientNet-B0: 92.34% Inception-V3: 93.65% DenseNet-121: 95.97%

CBAM-ResNet20 lightweight method surpassed the 96.81% accuracy achieved by the SE-ResNet50 model by implementing a dual attention module.

A comparative analysis of different DL models used to classify tomato leaf diseases within the PlantVillage dataset is presented in Table 4. This analysis aims to summarize their performance, highlights, and descriptions in disease recognition and classification.

6.2.2. Apple - PlantVillage

Many DL methods have been proposed in the past for recognizing apple leaf diseases. The PlantVillage apple sub-dataset is widely used for apple leaf disease recognition. This sub-dataset includes four classes: three apple leaf disease classes and one healthy class. Table 5 presents a comparative analysis of recently proposed DL approaches for recognizing apple leaf diseases within the PlantVillage apple sub-dataset. Francis and Deisy (2021) proposed a method that combines depth-wise and point-wise convolutions to reduce complexity and enable mobile compatibility. The resulting model, containing only 4.2 million parameters, achieved an impressive accuracy of 99.6%, making it suitable for mobile devices. The DenseNet-121 architecture was employed in Swaminathan et al. (2021) for the classification of multiple plant leaf diseases, achieving an accuracy of 99.21% for apple leaf diseases. The RGB color images were processed in HSV form to eliminate background information as part of the research work. Additionally, a web application was developed by the researchers.

A mask R-CNN with transfer learning techniques was used for apple leaf disease recognition, resulting in an accuracy of 99.1%, as demonstrated in Rehman et al. (2021). In addition, in this work, a hybrid contrast stretching technique was implemented to enhance the visual quality of the images. A multiscale stream approach was utilized to extract essential features from apple leaves, leading to enhanced performance compared to a single-stream network (Russel and Selvaraj, 2022). The model consists of 6 streams operating in three steps, with the input image size halved after each step. The lower scale (32×32) effectively learns the edges, contributing to an impressive accuracy of 98.92%. In Nandhini and Ashokkumar (2022), the DenseNet-121 framework was used, which incorporates the MHGSO algorithm to improve classification accuracy. However, an accuracy of 98.25% was achieved, which was lower compared to previous results. In this work, the transfer learning technique and MHGSO optimization were utilized to fine-tune hyperparameters during the optimization stage. In addition to these methods, a QBPSO-based approach was introduced that incorporates data augmentation, SqueezeNet for feature extraction, QBPSO for hyperparameter tuning, and LSTM for classification (Math et al., 2024). This method achieved a superior accuracy of 98. 76% compared to other techniques, including the discrete wavelet transformation (DWT), butterfly optimization algorithm (BOA) and chaotic salp swarm algorithm (CSSA).

6.2.3. Grapes - PlantVillage

In the PlantVillage dataset, the grape leaves sub-dataset consists of 4 disease classes: black rot, leaf blight, healthy and esca leaves. Using the transfer learning technique, the VGG models were used to recognize diseases in grape leaves, with the VGG-16 model achieving the highest accuracy of 98%, outperforming the VGG-19 model (Kavala and Pothuraju, 2022). For disease classification, the CNN classification model is

Table 5
A comparative analysis of various DL models on the same apple plant leaf diseases using the PlantVillage dataset is aimed at analyzing their performance, highlights, and descriptions in disease recognition and classification.

Authors	Architecture	Highlights	Description	Result
Francis and Deisy (2021)	CNN	Used convolution of depth-wise and point-wise methods, develop mobile version	The model was designed to minimize the number of parameters and computational cost in the framework. A CNN architecture with 12 layers and 4.2 million parameters was utilized. Batch normalization and ReLU functions were applied after the depth-wise and point-wise convolutions. Computational complexity was reduced, enabling the model to run faster.	Proposed model: 99.6% Inception-v3: 99.4% VGG: 99.5%
Swaminathan et al. (2021)	DenseNet-121	Remove the background features from the image using HSV form	The DenseNet-121 architecture was implemented with a transfer learning methodology in this research work. Further improvement for the feature extraction process was undertaken to eliminate the background features in the image dataset. The HSV form method was used to achieve this improvement. A total of 7.3125621 million parameters were used for training purposes.	Proposed model: 99.21% FSMDC: 94.8% IRPDD: 77.2% ACDMT: 85.4% MLPdd: 90.7%
Rehman et al. (2021)	MASK RCNN	Kapur's entropy and the MSVM (EaMSVM) approach for selecting strong features for the final classification.	The MASK RCNN model was employed with the ResNet101 as the base model. Kapur's entropy and the MSVM (EaMSVM) approach were utilized in the feature selection process to identify robust features for the final classification. The highest accuracy in the minimum time was achieved by the average pooling layer when comparing performance.	Proposed model: 99.1% VGG-16: 90.4%
Russel and Selvaraj (2022)	CNN	Six stream parallel learning using learnable and customized Law's mask filter.	The framework is designed to learn the inherent features of the dataset images. A six-stream parallel process is employed by the CNN model. In this process, the same input image size of 128×128 is reduced to 64×64 in the third and fourth streams and further reduced to 32×32 in the fifth and sixth streams. Law's Mask filter is also customized by the model.	Proposed model: 98.91% VGG-16: 90.4% VGG-19: 90% Inception-v3: 83% ResNet-50: 80%
Math et al. (2024)	SqueezeNet	The QBPSO-based LSTM dimensional approach enhances the detection of leaf disease	The QPBPSO approach was employed by the model for hyperparameter tuning and leaf disease classification using the QBPSO-based LSTM technique.SqueezeNet layers perform the feature extraction process.	Proposed model: 98.7% DWT: 98.6% BOA: 95.07%

Abbreviations: Feature Set based Multi-Diseases Classification (FSMDC), Image Recognition based Plant Disease Detection (IRPDD), Automated Crop Disease Management Techniques (ACDMT), Machine Learning-based Plant Disease Diagnosis (MLPDD), Evolutionary Algorithms for Multi-class Support Vector Machines(EaMSVM), Quantum-Behaved Particle Swarm Optimization (QBPSO), Long Term-Short Memory (LSTM), Discrete Wavelet Transformation (DWT), Butterfly Optimization Algorithm (BOA).

used in combination with the Improvised KNN (IKNN) model. The IKNN model was chosen for classification due to its efficient performance compared to CNN, which requires a longer training time. Consequently, the combined CNNC-IKNN model obtained the highest accuracy of 98.07%, exceeding the accuracy of the CNNC model without IKNN, which reached 96.60% (Shantkumari and Uma, 2022). Despite the slight difference in accuracy, it is important to note that while the VGG-16 model used its classifier, the CNNC model employed the IKNN approach. However, the VGG-16 model is considered the best option due to its ease of training (Kaur et al., 2022).

In Zhao et al. (2021), a CNN architecture utilizing transfer learning was implemented, incorporating the EfficientNet-B7 model, which attained a remarkable accuracy of 98.7%. This model showcased substantial improvements over existing CNN models. Generally, the attention module was found to be effective for feature extraction, and CNN was used for classification in the research. Specifically, grape leaves (99.24%) demonstrated higher accuracy compared to tomato leaves (96.81%), indicating superior performance within the grape leaves dataset. In another study conducted on the PlantVillage dataset, 8 different plant leaves, including grape leaves, were analyzed. The CNN architecture used a parallel multiscale technique, achieving an exceptional accuracy of 99.39% (Russel and Selvaraj, 2022).

6.2.4. Bell Pepper - PlantVillage

Due to the scarcity of Bell Pepper leaf disease data, there is a limited number of research articles have been published on the recognition of bell pepper leaf disease. Even within the PlantVillage dataset, the bell pepper plant sub-dataset includes only one disease class and healthy leaves. In ÖZCAN and DÖNMEZ (2021), a transfer learning approach with the DarkNet-19 model is used to recognize bell pepper leaf diseases. During the experimentation phase, the softmax classifier was replaced, and four alternatives were tested: Gaussian-based Naïve Bayes, linear SVM, KNN, and Decision Trees (DT). Among these, the SVM classifier achieved the highest accuracy of 98.8% and demonstrated faster processing times. In another study, a CNN architecture employing a parallel multiscale method achieved a higher accuracy of 99.465% in recognizing bell pepper leaf diseases, surpassing other models (Russel and Selvaraj, 2022).

6.2.5. Potato - PlantVillage

In the context of the PlantVillage dataset, three distinct categories of potato leaf diseases such as early-blight, late-blight, and healthy leaf were classified. Transfer learning techniques were utilized by Gardie et al. (2022) employing the Inception-V3 model, which resulted in notably superior performance compared to the AlexNet and GoogleNet models to recognize potato leaf diseases. An impressive accuracy rate of 98.3% was achieved through this approach. Particularly noteworthy was the capability of the model to achieve heightened accuracy within a condensed training period. Conversely, in Tiwari et al. (2020), a model similar to VGG-19 was developed through the application of transfer learning principles, incorporating various classifiers such as KNN, SVM, Neural Network, and Logistic Regression. The amalgamation of the VGG-19 model with Logistic Regression yielded an accuracy

Table 6
A comparative analysis of various DL models on the same tomato leaf diseases using the Taiwan Tomato dataset is conducted to analyze their performance, highlights, and descriptions in disease recognition and classification.

Authors	Architecture	Highlights	Description	Result
Liu et al. (2024)	ResNet	MTDL Framework Alternating training, weighting adjustment, and bidirectional knowledge Boost performance: attention mechanisms, binary Wavelet transform, and residual attention networks.	Methods called MTDL and DTF-KD are proposed in this study to streamline learning by minimizing dependence on teacher models. Enhanced performance and reduced model size in diagnosing tomato diseases are demonstrated through experiments using diverse architectures for teacher and student models.	Proposed model: 98.92% DHBP: 98.39% KD: 97.30% DKD: 97.56% AT: 97.39%
Ouamane et al. (2024)	HOWSVD-MDA	Two phase execution: HOWSVD approach and Culminates in MDA	The Tensor Subspace Learning utilized by HOWSVD and MDA is methodically executed in two phases. A pre-trained CNN model is employed to enhance the features.	Proposed model: 98.39% SNBLF: 96.14% Lw-Resnet-20-CBAM:88.92%
Linfeng et al. (2024)	Shared-twin neural network	Lightweight DL algorithm Shared-twin neural network employed on work	Shared-Twin Neural Networks are performed well in limited resource environments for fast identification, It is a Lightweight approach with a CNN base model.	Proposed model: 97.40% MobileNet-v2: 84.30% EfficientNet-v2: 70.40%
Thuseethan et al. (2022)	Siamese network with CNN	Novel Siamese network-based lightweight framework Backbone of the framework: CNN Improve performance: Attention schema	The feature embeddings of the two inputs are compared to measure their similarity in the proposed framework, which is used in the decision-making process with contrastive loss calculation, utilizing two sub-networks simultaneously.	Proposed model: 95.48% MobileNet-v2: 82.36% ShuffleNet-v2: 72.31% SqeezeNet-11: 66.22% Inception-v3: 68.47%

Abbreviations: Multi-Task Distillation Learning (MTDL), Decoupled Teacher-Free Knowledge Distillation(DTF-KD), Dual-stream Hierarchical Bi-linear Pooling (DHBP), Frequency Decomposed Local Ternary Pattern (FDLTP) Scale-Invariant Feature Transform (SIFT) Multi-Layer Perceptron (MLP), Knowledge Distillation (KD), Decouple Knowledge Distillation (DKD), Attention transfer (AT), Siamese Network-based Lightweight Framework(SNBLF) Scale-Invariant Feature Transform (SIFT), Isolation Forest (IF), Lightweight ResNet20 base CBAM (Lw-Resnet-20-CBAM), Higher-Order Whitened Singular Value Decomposition(HOWSVD), Multilinear Discriminant Analysis (MDA).

rate of 97.8%, surpassing the performance of both the VGG-16 and InceptionV3 models. However, it is imperative to acknowledge that the InceptionV3 model, utilized in the first-mentioned research study with a transfer learning approach but without the utilization of any specific classifier, demonstrated superior accuracy compared to the latter model. In Gupta et al. (2024a), utilized PLD dataset along with PlantVillage potato data to introduce PotCapsNet, which combines parallel atrous convolution, attention modules, and capsule networks for early disease classification. Validated on PlantVillage and PLD datasets, it achieved 97.81% and 97.95% accuracy, surpassing existing methods.

6.2.6. Tomato - Taiwan

The Taiwan tomato leaf disease dataset has attracted significant interest, leading to the publication of numerous works. A comparative analysis of various ML and DL models proposed for recognizing tomato leaf diseases on the Taiwan tomato leaf dataset is presented in Table 6. Liu et al. (2024) employed the MTDL framework integrated with a binary wavelet attention mechanism and a residual attention network. This approach achieved outstanding performance, with an accuracy of 98.92%. In addition, Ouamane et al. (2024) suggested a novel approach utilizing CNN models in tensor subspace learning, namely Higher-Order Whitened Singular Value Decomposition (HOWSVD). This approach enhanced the system's discriminatory power and achieved remarkable classification accuracy rates of 98.39% on the Taiwan dataset. Their work set a new benchmark in tomato disease recognition, demonstrating state-of-the-art performance. In another work, the utilization of the MTLD framework with ResNet and other state-of-the-art models has been explored to enhance performance (Liu et al., 2024). Thuseethan et al. (2022) proposed a novel lightweight Siamese network for tomato disease recognition, demonstrating effectiveness with small and imbalanced datasets, and achieving an accuracy of 95.48

6.2.7. Citrus

A comprehensive analysis of various citrus leaf disease recognition approaches tested on the citrus dataset is presented in Table 7. The majority of these works used CNN-based architectures for effectively classifying tomato leaf diseases. Lanjewar and Parab (2024) deployed a CNN model with transfer learning, using Adam optimizers on an augmented dataset, in a Platform as a Service (PaaS) environment tailored for mobile systems. The CNN model, designed for efficient mobile deployment, achieved an impressive classification accuracy of 99%. In Khattak et al. (2021), a CNN model is presented for recognizing citrus diseases, while in Elaraby et al. (2022), AlexNet and VGG19 are used as feature extractors along with geometrical training data augmentation. In another study, simplified variants of the MobileNetV2 architecture were used as embedding modules within the framework, achieving an accuracy of 97.10% on the citrus dataset (Janarthan et al., 2022). Lanjewar and Parab (2024) employed DL models such as ResNet152V2, InceptionResNetV2, DenseNet121, and DenseNet201 to develop a cloud-based service that efficiently handles data processing and model deployment. Their study illustrates the effective integration of advanced neural networks into cloud infrastructure, demonstrating the versatility and robustness of these models in practical applications.

6.3. Specialized approaches

In recent years, specialized methods for agricultural analysis have significantly advanced, driven by the integration of state-of-the-art ML and image-processing technologies. These innovations have enabled precise assessments in various domains. This subsection explores four popular specialized approaches used for automated plant leaf disease recognition: transfer Learning Approach, attention methods, federated learning, explainable artificial intelligence, and quantum deep learning.

6.3.1. Transfer learning approach

Transfer learning is widely recognized as an effective approach in DL experiments, where a model trained for one task is reused for a related task. Instead of starting the training process from scratch, the knowledge gained from solving one problem is leveraged and applied to a different but related problem. In the pre-training phase, a DL model is utilized on a large dataset, such as ImageNet with 1.2 million images. The transfer learning approach is illustrated in Fig. 10, has been utilized in several recent studies. For example, in Khan et al. (2022), four pre-trained models were evaluated and the best performing model was selected based on its accuracy (Khan et al., 2022). The main advantages of using transfer learning of DL models include a reduced training time and lower computational power requirements (Hussain et al., 2018). A major challenge in DL is the need for large datasets; however, high

Table 7
A comparative analysis is being conducted on various DL models for the same citrus leaf diseases using the citrus leaf dataset, to analyze their performance, highlight their characteristics, and describe their roles in disease recognition and classification.

Authors	Architecture	Highlights	Description	Result
Lanjewar and Parab (2024)	CNN	CNN-Adam-Aug with transfer learning was deployed on a PaaS mobile system.	Transfer learning and Adam optimizers were utilized on an augmented dataset by the CNN architecture. Furthermore, a compact CNN model was implemented on a Platform as a Service (PaaS) cloud infrastructure tailored for a mobile-oriented system.	Proposed model: 99% DenseNet-201-Aug: 94% DenseNet-121-Aug: 90%
Janarthan et al. (2022)	Siamese network with CNN	Embedding module: Siamese network Distance-based interpretation Outcome: Mobile application	An online augmentation technique was applied to expand the training samples. Embeddings were created for patches of each class and clustered. The trained embedding module, along with the class prototypes, was used to build the final classification network.	Proposed model: 97.10% MobileNet-v2: 94.60%
Janarthan et al. (2020)	Siamese network with DCNN	Patch-based classification network Use embedding model (K-Means clustering)	In the pre-processing step, each whole leaf is divided into patches and embeddings are calculated using a pre-trained embedding module. The K-Means clustering algorithm is applied to cluster the computed embeddings.	Proposed model: 95.04% Inception-v3: 94.79% DenseNet-201: 94.80% VGG-16: 92.93% MobileNet-v2: 94.37%
Khattak et al. (2021)	CNN	Keras image preprocessing Activation: Softmax	Image preprocessing in Keras is performed using the Image-Data-Generator class and API, which allows normalization, pixel scaling, and data augmentation. Feature maps are obtained through convolution operations.	Proposed model: 94.55% MobileNet-v2:87.28%
Elaraby et al. (2022)	AlexNet With SGDM	Feature extraction: AlexNet and VGG19. Geomtry data augmentation	GDM optimizers were employed in conjunction with transfer learning using DL models AlexNet and VGG19 Data augmentation techniques were applied to mitigate overfitting issues during the model training phase.	Proposed model: 94.3% VGG-16: 92.9%

Abbreviations: Gravitational Search Algorithm (GSA).

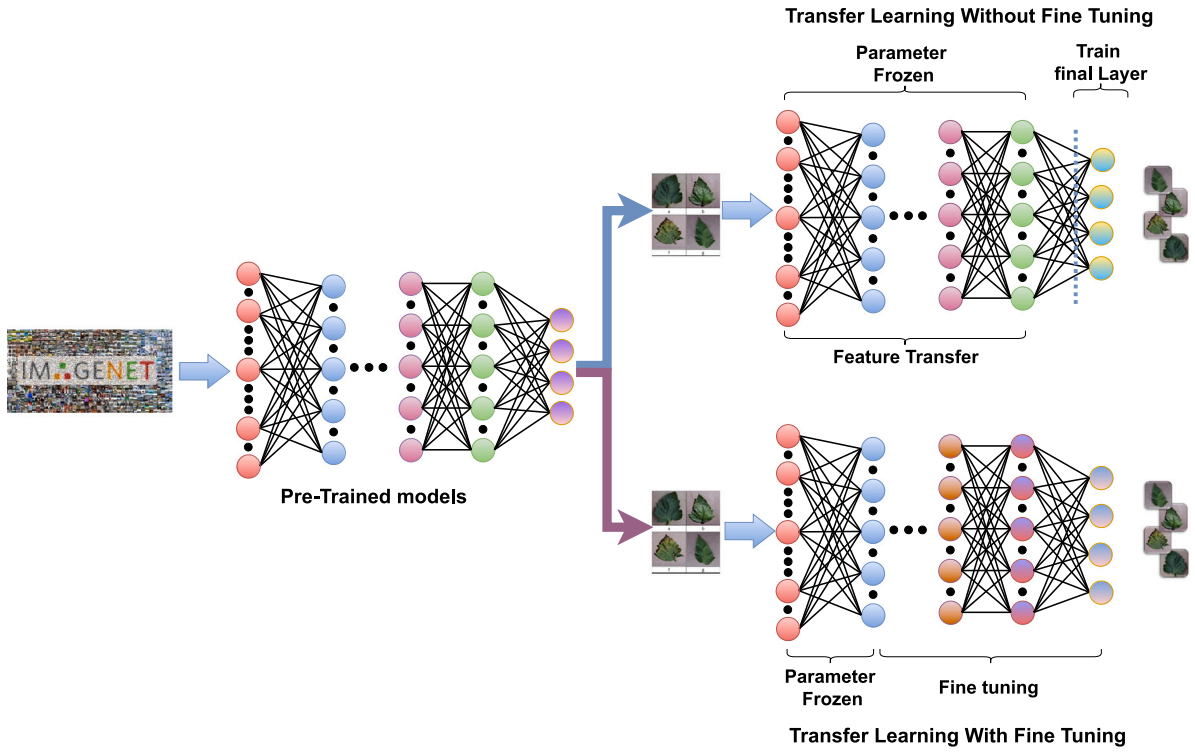


Fig. 10. The general transfer learning approaches employed in plant disease recognition, leveraging pre-trained models and fine-tuning them on plant disease datasets.

accuracy can still be achieved with smaller datasets through transfer learning, making this approach highly valuable (Shrivastava et al., 2019; Coulibaly et al., 2019).

Similar to other domains, transfer learning of deep models showed significant improvements in plant leaf disease recognition as well. Two transfer learning strategies are implemented in Ahmad et al. (2020): (1) feature extraction only (no fine-tuning) and (2) feature extraction and fine-tuning of high-dimensional layers' parameters. Among them, fine-tuning proved to be more effective, with the best performance demonstrated by Inception-v3 among the DL models used in this study, which included VGG-16, VGG-19, ResNet, and Inception-v3 (Ahmad et al., 2020). In Liu et al. (2023), the selective kernel MobileNet model, a lightweight DL model, was applied with the transfer learning technique to classify three types of crops—Bell Pepper, Potato, and Tomato—demonstrating high effectiveness with an impressive accuracy of 99.28%. Not only in plant leaf disease recognition, but it has also demonstrated state-of-the-art performance in other precision agriculture areas such as pest detection. In the citrus pest detection task,

a lightweight transfer learning model named MobileNet-v3 + RPBM achieved an accuracy of 91%, while the experiment also evaluated three other models: GoogLeNet, ResNet, and VGGNet (Wang et al., 2023). Among these models, MobileNet-v3 + RPBM was selected as the most appropriate for the pest detection task with the highest accuracy.

6.3.2. Attentions methods

In recent years, attention networks have gained widespread use in various fields, including machine translation and generative adversarial networks, due to their ability to extract discriminative features from areas of interest. This makes them especially valuable for plant leaf disease detection, where accurately identifying and focusing on affected regions is crucial for effective diagnosis (Zhao et al., 2021). Zhao et al. (2022) proposed a modified CBAM to be integrated within the RI-Net network, aiming to improve the classification of plant leaf diseases by more precisely locating disease spots in images and enhancing the model's recognition accuracy. Their model demonstrated higher running efficiency and achieved a recognition accuracy of 99.55%, outperforming Inception-v3, Xception, ResNet-50, and Inception-ResNet-v2. The CBAM attention module improves the detection of subtle lesion features in the passage layer by extracting both channel and spatial information, thereby minimizing feature redundancy (Sharma et al., 2023a).

Bhujel et al. (2022) developed a lightweight convolutional neural network using ResNet topology by incorporating various attention modules, including the CBAM, SE Attention Module, Self-Attention (SA) Module, and Dual Attention (DA) Module. Their study emphasized that the SE module is the lightest and most efficient attention module, significantly enhancing model performance. Another study introduced an AAR network to address CNN locality constraints and limited training samples. The AAR network, which integrates residual blocks and attention modules, and employs a CVGAN for augmentation, achieved 97.04% accuracy without augmentation, 98.91% with data generation and augmentation, and 99.03% with data augmentation alone (Abawatew et al., 2021). Zhang et al. (2024a) developed the frequency-domain attention network (FdaNet) to adjust feature importance across different frequency domains. They also proposed the hybrid attention network (HaNet) to focus on multi-dimensional feature information, achieving 98.83% and 98.77% recognition accuracy on 50-layer and 101-layer networks, respectively, in citrus disease detection.

6.3.3. Federated learning approach

In real-world scenarios, the identification of plant diseases is often constrained by the challenge of obtaining sufficient samples for all disease classes from a single location. Nonetheless, the number of classes with adequate samples can be increased by exploring diverse geographical regions. However, a significant challenge is presented by the integration of datasets from various locations. This issue is addressed by the federated learning (FL) approach, which enables the model to be trained locally on different resource-constrained edge devices using small datasets from various locations. Subsequently, the knowledge is aggregated into a global model, which can then be utilized across diverse geographical locations. FL is currently experiencing heightened demand due to its ability to address critical challenges such as limited data availability across multiple locations, security concerns, access permissions, data confidentiality, and the limitations of resource-constrained edge devices. Recent works have explored various types of FL, including centralized FL, decentralized FL, cross-device FL, cross-silo FL, horizontal FL, and vertical FL. In the future, it would be valuable to conduct a comprehensive practical analysis of these different FL architectures, along with an in-depth investigation into privacy-preserving techniques within federated machine learning. Furthermore, the characteristic of lightweight DL frameworks has played a significant role in shaping the design of federated learning architectures (Huang and Chang, 2020; Sethy, 2020; Mensah et al., 2023; Thite et al., 2024). Mensah et al. (2023) employed a federated learning approach in the DL training phase for classifying rice-leaf diseases. Their study found that EfficientNet-b3 and MobileNet-v2 outperformed other models, achieving a validation accuracy of over 99% for both Independent and Identically Distributed (IID) and non-IID datasets. In another investigation focusing on rice leaf disease recognition using DL and federated learning techniques, EfficientNet-b3 consistently achieved 99% accuracy during both training and evaluation phases. Notably, among the models assessed in the study, EfficientNet-b3 was identified as a lightweight model. During the training phase, ten clients were involved, each with varying quantities of IID and non-IID data samples, some of which were limited in number (Huang and Chang, 2020).

6.3.4. Explainable artificial intelligence

Explainable Artificial Intelligence (XAI) aims to mitigate the opacity of DL algorithms, which often operate as black boxes, making predictions difficult to interpret. This opacity can lead to unintended biases and decreased trust, impacting feature optimization and accuracy. Researchers are developing methods to elucidate these models, collectively referred to as XAI, with the primary objective of enhancing interpretability and fostering trust, fairness, and accountability in AI systems (Sagar et al., 2023; Nigar et al., 2024; Natarajan et al., 2024).

Numerous methods for explaining model predictions include SHAP (Shapley Additive Explanations), which employs a game theory-based framework for consistent, model-agnostic explanations. LIME (Local Interpretable Model-agnostic Explanations) offers local approximations with interpretable models. Grad-CAM enhances convolutional neural networks (CNNs) interpretability by highlighting critical image regions using class gradients. Microsoft's InterpretML combines techniques like SHAP, LIME, and partial dependence plots to elucidate feature importance and prediction behaviors.

Natarajan et al. (2024) utilized the EfficientNetB0 model within the LIME framework of XAI, achieving an accuracy of 99.69% across 38 plant diseases. A transfer learning approach was applied using three pre-trained models, along with the LIME technique, to evaluate their performance. The highest-performing model, EfficientNetV2L, attained an accuracy of 99.63% when tested on the same set of 38 plant diseases, while the other models assessed were MobileNetV2 and ResNet152V2 (Mehedi et al., 2022). Prince et al. (2024) focused on a lightweight deep learning framework, employing a hybrid CNN-SVM model to classify four plant diseases, achieving an accuracy of 99.09%. They also applied the Grad-CAM XAI technique to improve the interpretability of the model in the disease classification process.

6.3.5. Quantum deep learning

Recent advancements in Quantum Deep Learning (QDL) offer significant improvements in plant disease detection. By combining quantum computing with DL, models like Quantum Convolutional Neural Networks (QCNNs) have enhanced accuracy, achieving 96.9% in classifying leaf blight (Genemo, 2023). A hybrid quantum ResNet18 model further outperformed traditional methods, with 87% validation accuracy on large datasets like ImageNet (Kujani et al., 2024). The use of quantum particles in training also reduced energy consumption and processing costs, demonstrating the potential of QDL in resource-constrained agricultural settings (Padmanabhuni et al., 2023).

Sowmiya and Krishnaveni (2023) present the Improved Quantum Whale Optimization with Principal Component Analysis (IQWO-PCA) to enhance machine learning models for tomato disease classification using the Plant Village dataset. This approach employs transfer learning and optimized hyperparameters across four deep neural networks (AlexNet, VGG16, ResNet50, DenseNet121), resulting in improved classification accuracy and reduced loss. The hybrid system surpasses traditional machine learning methods, indicating its potential for proactive agricultural disease management.

Additionally, research on tomato leaf diseases using quantum machine learning techniques, including Quantum Neural Networks (QNN)

and Quantum–Classical Hybrid Convolutional Neural Networks (QCHCNN), achieved detection accuracies of 87%, 90%, and 99%, with QCHCNN being favored for its cost-effectiveness and rapid detection, underscoring the applicability of quantum deep learning in agriculture (Sudha and Vignesh, 2024). Önler (2023) compared a MobileNet v2 model with a hybrid classical–quantum network for leaf disease classification, finding accuracies of 99.25% and 92.64%, respectively, with potential generalization issues in the hybrid model. Singh and Khan (2023) found classical models (91%) outperforming quantum models (72%) for plant disease detection, highlighting future work on spiking neural networks and quantum hardware.

These specialized approaches offer promising advancements in automated plant leaf disease recognition. Leveraging advanced focus techniques allows models to concentrate on key features, improving accuracy in identifying diseases. These progress will empower farmers with more efficient tools for crop disease management, ultimately contributing to increased agricultural productivity and food security.

7. Applications

Applications for plant leaf disease detection models are extensive and possess substantial potential for improving agricultural practices. These models, which utilize advanced ML and DL algorithms, are capable of accurately identifying and diagnosing various plant diseases from leaf images. This capability is essential for facilitating real-time and precise interventions, enabling farmers to promptly address disease outbreaks and avert extensive crop damage (Khattak et al., 2021). Notable applications include:

1. Mobile Applications for Farmers: To facilitate real-time disease detection through smartphone cameras, providing immediate feedback and treatment recommendations. These applications feature a user-friendly interface, ensuring ease of use for farmers (Janarthan et al., 2022; Verma et al., 2019; Khan et al., 2023).
2. Web-based Applications: To provide accessible platforms for farmers, serving as convenient online hubs. These platforms offer a user-friendly interface for accessing essential agricultural information and resources related to automated methods for identifying and classifying plant leaf diseases (Pendam et al., 2023; Islam et al., 2023).
3. Robotic Applications: Enhance agricultural practices by implementing precision irrigation systems that utilize disease detection data, ensuring targeted and efficient water distribution. Additionally, these systems support automated treatment applications and optimize resource management, leading to improved crop health and reduced wastage (Abed et al., 2021; Forhad et al., 2023).
4. Drone-based Monitoring Systems, as a vital tool in large-scale crop surveillance, enable early disease detection through high-resolution imaging and data collection from aerial perspectives. These systems utilize sophisticated sensors and imaging technologies to capture multispectral and hyperspectral images, revealing disease symptoms not visible to the naked eye. Automated analysis through ML algorithms facilitates timely health status reporting, aiding in efficient disease mitigation strategies and reducing crop losses significantly (Albattah et al., 2022; Abbas et al., 2023; Chin et al., 2023).
5. Automated Irrigation and Treatment Systems: These are advanced agricultural practices that integrate precision irrigation techniques with data from disease detection mechanisms. These systems, equipped with image analysis capabilities and the ability to fuse data from multiple sources, enable the automated application of treatments such as fertilizers, pesticides, or water based on real-time data insights. By employing efficient resource management strategies, they aim to optimize water usage, minimize wastage, and enhance crop health. This approach not only improves agricultural productivity but also contributes to reducing the environmental impact associated with conventional farming practices (Fatima et al., 2021).
6. Precision Agriculture Platforms: These platforms encompass sophisticated analytical tools that provide in-depth insights into crop health and disease patterns, leveraging real-time data analytics and sophisticated monitoring tools. By harnessing large volumes of agricultural data, these platforms enable early detection of diseases, precise treatment application, optimized resource allocation, and proactive management strategies. This holistic approach not only improves operational efficiency but also contributes to sustainable agricultural practices, ultimately leading to increased productivity, minimized environmental impact, and empowered farmers with data-driven decision-making capabilities for superior crop management and disease mitigation (Sishodia et al., 2020).
7. Educational Tools for Agricultural Training: Such training modules on disease identification for farmers and students, interactive simulations, and case studies, along with improved dissemination of best practices, play a vital role in integrating plant disease detection and mitigation practices. By incorporating interactive modules on plant health monitoring and disease identification into these tools, farmers and agricultural professionals can enhance their understanding of various crop diseases and their management strategies (Astani et al., 2022).
8. Research and Development: Combining research and development with plant disease detection and mitigation involves advancing the understanding of diseases through research and translating findings into practical solutions, such as developing accurate detection techniques and innovative mitigation strategies. This includes data collection for developing disease-resistant crops (Liu et al., 2024), conducting field trials in collaboration with research institutions, and continuously improving detection algorithms. These efforts lead to improved crop health, productivity, and sustainable farming practices.

Integrating advanced ML and DL models into agricultural applications can significantly benefit farmers by enabling more precise and timely disease detection and management, thereby achieving higher productivity. This integration aids in reducing crop losses by enabling early identification and treatment of plant diseases, ultimately leading to improved crop yields.

8. Limitations and challenges

In the field of plant disease detection, several significant limitations impede the creation of effective and reliable models. Addressing these limitations is essential to enhance the accuracy and applicability of disease detection systems in real-world agricultural scenarios.

8.1. Dataset-related

When research or analysis is conducted, the inherent limitations of the utilized dataset must be acknowledged. These limitations should be recognized to contextualize the findings, ensure accurate interpretation of results, and highlight areas requiring caution. Dataset limitations can stem from various factors, such as data collection methods, the scope of the data, and potential biases. Understanding these constraints allows for a critical assessment of the robustness and generalizability of the conclusions.

1. **Small sample size of available datasets:** The sample size of available datasets is often limited, which poses a significant challenge. The full diversity of real-life scenarios is not captured by these small datasets, leading to models that may not generalize well to different conditions or varieties of plants (Khattak et al.,

2021). The risk of overfitting is also increased by the limited data, where specific details and noise of the training data are learned by models rather than general patterns, further reducing their effectiveness in practical applications.

2. **Imbalanced data:** Another popular limitation is the presence of imbalanced data, which exacerbates the challenge, as certain disease classes may be underrepresented compared to others in the dataset. The model learning process can be skewed by this imbalance, leading to biased predictions and reduced performance, particularly for minority classes (Thuseethan et al., 2022). Addressing this issue in predictive modeling often demands specialized techniques, such as re-sampling methods, as classification algorithms typically assume a balanced distribution of examples across all classes.
3. **In-field applicability issues:** Despite the encouraging outcomes observed in controlled settings, numerous models remain largely untested in practical agricultural contexts. Real-world agricultural scenarios exhibit a wide range of disease classes, while datasets often provide limited and diverse classes. This disparity between the complexity of real-world conditions and the simplification of datasets can result in inadequate assessment of disease severity, which can lead to misdiagnoses and inappropriate treatment recommendations (Liu et al., 2024).
4. **Environmental and image quality issues:** The challenges encompassing noise, background clutter, shadows cast on the leaves, low image quality, the coexistence of multiple leaves, diverse textures, and brightness variations have been identified (Astani et al., 2022; Saraswathi and FarithaBanu, 2023). UAV-captured images often face limitations such as adverse weather conditions, including fog, and low resolution, which can hinder effective analysis. However, advanced techniques and devices have been developed to address these challenges. For instance, an IoT–Fog robotic system utilizing deep learning and de-hazing techniques has been proposed to classify crops under hazy conditions (Kansal et al., 2023). The use of high-quality cameras and sensors, such as RGB, multispectral, hyperspectral, thermal cameras, and laser scanners, significantly enhances image quality when integrated with UAV systems. Additionally, advanced preprocessing techniques further improve image clarity, enabling more reliable and accurate analyses (Zhu et al., 2024).
5. **Uniform backgrounds in datasets:** In some instances, datasets were acquired from a homogeneous background, presenting a notable challenge. Models trained solely on such homogeneous background datasets often encounter difficulties when confronted with data featuring intricate backgrounds in real-time contexts. This constraint impedes the capacity of the model for generalization and its efficacy in varied, real-world settings (Zhang et al., 2024a). The applicability of these models in real-world field conditions remains unexplored, representing a critical aspect for assessing their practical effectiveness.

8.2. Model-related

Traditional DL models for the detection of plant diseases often encounter substantial challenges associated with extended training times and limited resources. These constraints impede their development and implementation, especially in mobile-based applications and resource-constrained settings. Although lightweight models offer partial solutions to these issues, it is imperative to comprehend the specific limitations of conventional models to enhance their practical applicability (Saraswathi and FarithaBanu, 2023).

1. **Training time:** The intricate architectures and extensive parameters of DL models require substantial data processing, thus extending the training period. This prolonged duration can hinder model development and complicate iterative enhancements (Saraswathi and FarithaBanu, 2023; Janarthan et al., 2022).
2. **Resource constraints:** DL methods demand substantial memory and processing power, which are often beyond the capabilities of many devices, especially mobile devices. This limitation is particularly significant for mobile-based applications, as smartphones typically lack the necessary hardware to support intensive computation and memory usage. Consequently, DL models are less feasible for direct operation on mobile devices, limiting their utility in resource-constrained environments. Addressing these resource constraints is essential to making DL models more accessible and effective in practical applications (Janarthan et al., 2022; Yu et al., 2022).
3. **Overfitting and underfitting:** Learning models are susceptible to overfitting and underfitting during training due to their inherent complexity. Underfitting occurs when a model fails to capture the underlying patterns, leading to high bias and low variance, often due to insufficient data or overly simplistic models. Overfitting, on the other hand, arises when a model learns too much from the training data, including noise, resulting in low bias but high variance. This phenomenon is particularly common in flexible, non-linear models. Although it is challenging to completely eliminate these issues, careful model tuning and validation can mitigate their impact. Al-Gaashani et al. (2024) demonstrated this by observing overfitting in their convolutional neural network during the initial training epochs.
4. **Vanishing gradient problem:** The vanishing gradient problem significantly impacts the training of deep neural networks, stemming primarily from the model architecture rather than the data. This issue arises when gradients become exceedingly small during back-propagation, particularly in deep networks with numerous layers. Consequently, this phenomenon impedes effective training by decelerating weight updates in the initial layers, resulting in slow convergence or learning stagnation. To mitigate this problem and enhance the efficiency of training deep neural networks, various techniques are employed, including appropriate weight initialization, the selection of activation functions, and the application of regularization methods (Orchi et al., 2021).

Therefore, addressing these limitations is crucial for the successful deployment of DL models in resource-constrained environments. Overcoming these constraints ensures that DL models can operate efficiently and effectively, maintaining high performance without exceeding the available resources. Table 8 further presents a comprehensive overview of the limitations associated with various DL methods used for plant disease classification. It offers a comparison of plant disease detection approaches in recent research, highlighting key challenges such as data imbalance, resource constraints, complex backgrounds, infield applicability, small data sets, and cross-database compatibility.

9. Recommendations and future directions

To improve the performance of DL models for the detection of plant disease, it is essential to address several key challenges outlined in the previous section, including class imbalance, insufficient data, infield applicability, and computational constraints. Implementing targeted strategies to tackle these issues can significantly improve the accuracy, generalizability, and efficiency of the models, thereby increasing their effectiveness in real-world applications.

1. **Addressing class imbalance and insufficient data:** Addressing class imbalance and insufficient data across regions is essential for enhancing model performance. This can be achieved by incorporating datasets of varying sizes into the training process

Table 8
Limitations of various DL methods for plant disease classification.

Authors	Imbalance data	Resource constraint	Complex background	Infield applicable	Small data	Cross database
Janarthan et al. (2020)	✱	✓	✱	✗	✗	✗
Fuentes et al. (2021)	✗	✗	✗	✱	✗	✓
Sunil et al. (2021)	✗	✗	✓	✱	✗	✓
Astani et al. (2022)	✱	✗	✱	✱	✗	✗
Yu et al. (2022)	✱	✱	✗	✗	✗	✗
Thuseethan et al. (2022)	✱	✓	✗	✓	✓	✗
Bhujel et al. (2022)	✱	✓	✗	✱	✗	✗
Cheng et al. (2022)	✱	✗	✗	✓	✗	✗
Janarthan et al. (2022)	✓	✓	✗	✓	✱	✗
Çetiner (2022)	✓	✱	✱	✗	✗	✗
Ahmad et al. (2023a)	✱	✗	✱	✓	✗	✓
Islam et al. (2023)	✱	✗	✗	✱	✗	✗
Ngugi et al. (2023)	✱	✱	✓	✱	✗	✗
Wu et al. (2023)	✗	✓	✓	✱	✗	✗
Lanjewar and Parab (2024)	✓	✓	✗	✓	✗	✗
Linfeng et al. (2024)	✱	✓	✗	✱	✗	✗

✓- Addressed and resolved, ✗- Remains unaddressed and unresolved, ✱ - Partially resolved.

and utilizing transfer learning techniques (Khattak et al., 2021). Additionally, novel augmentation methods, such as Generative Adversarial Networks (GANs), can further enhance the diversity and realism of training data, thereby improving model performance in real-world applications. Furthermore, various preprocessing techniques, such as geometric transformations or intensity transformations, can be employed to enhance the quality of the data.

2. **Improving infield applicability:** To enhance the model's generalization and performance in diverse real-world scenarios, it is crucial to collect new datasets that encompass images with varied and complex backgrounds (Cheng et al., 2022). This strategy ensures that models are trained on data that better represent real-world conditions, thus improving their applicability in practical settings (Zhang et al., 2024a). Farmers are increasingly adopting cutting-edge technologies in smart farming, with a focus on early disease detection and enhancing agricultural processes more efficiently, all while promoting sustainable development (Gupta and Tripathi, 2024; Sharma et al., 2022). Moreover, smart vehicles powered by 6G technology are transforming the farming sector by integrating advanced IoT with night vision systems, enabling enhanced visibility and safety in low-light conditions (Gupta et al., 2024b).
3. **Resolving image quality issues:** Applying preprocessing techniques, such as noise reduction, contrast enhancement, and normalization, can significantly improve image quality (Khattak et al., 2021). Noise can be reduced using techniques, such as the Gaussian filter (Balaji et al., 2023) or median filtering, while histogram equalization (Çetiner, 2022; Astani et al., 2022) can enhance contrast and highlight disease features. Additionally, employing data augmentation techniques that simulate various lighting conditions, shadows, and backgrounds can increase model robustness to variations in image quality (Al-Gaashani et al., 2024). Methods such as random cropping, rotation, and brightness adjustments are particularly effective (Janarthan et al., 2022).
4. **Enhancing model efficiency with lightweight models:** Techniques such as network parameter quantization and pruning can be employed to compress deep models (Han et al., 2015). Additionally, more efficient neural network architectures, such as MobileNets and SqueezeNet, can be utilized to reduce computational load and memory usage while maintaining performance (Janarthan et al., 2020). Sharma et al. (2023b) examined deeper lightweight CNNs designed to mitigate vanishing gradient problems while employing point-wise and separable convolutions for efficient feature extraction. Leveraging edge computing can further offload processing tasks from mobile devices to nearby edge servers, thus reducing the burden on the mobile device itself. By adopting these strategies, the limitations related to training time and resource constraints can be significantly mitigated, enhancing the feasibility of deploying DL models in mobile and resource-constrained environments.
5. **Lack of cross-dataset validation:** The lack of cross-dataset validation experiments is a significant limitation in the generalization of ML models. Cross-dataset validation is crucial for ensuring that models can adapt effectively to new, previously unseen data drawn from the same distribution as the data used during model development. Without this type of validation, the robustness and versatility of the models remain uncertain, potentially leading to overfitting to specific datasets and poor performance in real-world scenarios. From this survey, it is evident that future researchers should prioritize conducting cross-dataset validation experiments. This practice will enhance the reliability of the models and ensure their applicability across diverse datasets, ultimately advancing the field and improving the practical utility of ML models.

By emphasizing these aspects, along with the integration of emerging technologies such as the IoT, Explainable AI, Federated Learning, Quantum Deep Learning, and Smart Farming, future research can substantially enhance the accuracy, scalability, and practical applicability of automated plant disease recognition systems in real-world agricultural settings.

10. Conclusion

Agriculture plays a vital role in sustaining human civilization, but it faces numerous challenges posed by various factors. The emergence of infectious plant diseases remains a primary threat, necessitating automated agricultural practices and innovative solutions. Adhering to the PRISMA guidelines, this survey methodically highlights the common stress factors affecting plant health and productivity, offers a comparative analysis of various plant disease datasets, emphasizing their sizes, diversities, and qualities, while thoroughly examining and comparing the effectiveness of state-of-the-art DL methods for plant leaf disease recognition and diagnosis. It discusses the limitations of existing approaches and provides actionable recommendations for enhancing their real-world applicability. By analyzing diverse datasets and their technical settings, the study offers valuable insights into the methodologies used for dataset collection, the diversity of plant species covered, and the range of disease categories addressed. This study highlights a significant research gap in the need for enhanced lightweight models suitable for real-world applications and also emphasizes the shift from traditional DL models used with laboratory datasets. It explores the advantages of DL techniques, supported by extensive datasets and

advanced computational capabilities, and discusses transfer learning and comparative analyses of DL-based methods. Furthermore, the survey emphasizes the potential of plant leaf disease detection models to improve agricultural practices by enabling real-time and precise interventions to prevent extensive crop damage, while also addressing the importance of real-world datasets, ideal dataset concepts, identified limitations, and providing recommendations and future directions for advancing plant disease recognition. Despite the advancements, several challenges make it difficult to create effective and reliable models for plant disease detection. This survey addresses such limitations to enhance the accuracy and applicability of disease detection systems in real-world agricultural scenarios, providing a foundation for future research and highlighting the need for innovative solutions to meet the growing demand for food and mitigate the impacts of plant diseases on global agriculture.

CRediT authorship contribution statement

Romiyal George: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Selvarajah Thuseethan:** Writing – review & editing, Validation, Supervision, Investigation, Formal analysis, Conceptualization. **Roshan G. Ragel:** Writing – review & editing, Validation, Supervision, Investigation, Formal analysis, Conceptualization. **Kayathiri Mahendrakumaran:** Validation, Investigation, Formal analysis, Conceptualization. **Sivaraj Nimishan:** Validation, Investigation, Formal analysis, Conceptualization. **Chathrie Wimalasooriya:** Validation, Investigation, Formal analysis, Conceptualization. **Mamoun Alazab:** Validation, Investigation, Formal analysis, Conceptualization.

Ethical approval

No ethical approvals were sought as only publicly available data were used in this study.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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